

E3AN



DEPARTMENT OF
ELECTRONICS & COMMUNICATION
ENGINEERING

ASANSOL ENGINEERING COLLEGE

**Project on
E-Waste Material**

ROBOT E3AN



**DEPARTMENT OF ELECTRONICS AND
COMMUNICATION ENGINEERING
ASANSOL ENGINEERING COLLEGE**
Vivekananda Sarani, Kanyapur, Asansol, West Bengal – 713305

CERTIFICATE

This is to certify that the project entitled 'E3AN' developed by Team Matrix is based upon their own work carried out in Asansol Engineering College, Department of Electronic & Communication Engineering under the affiliation of Maulana Abul Kalam Azad University of Technology, West Bengal. This project is undertaken as part of the preparations for the NAAC Accreditation.

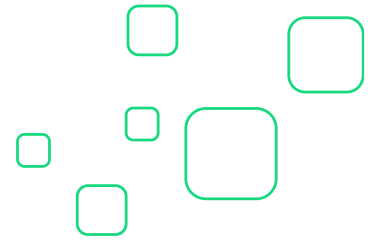
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Message From The Project Supervisor



Dear Team E3AN,

I am delighted to convey my heartfelt congratulations and appreciation to each member of the E-Waste Robot Project team for the exemplary work done on the "E3AN: Transformative E-Waste Repurposing" project. Your dedication, ingenuity, and collaborative spirit have resulted in a remarkable achievement that not only aligns with our academic objectives but also exemplifies the spirit of innovation and sustainability.

Throughout the journey of this project, it has been inspiring to witness your team's commitment to pushing the boundaries of technology while addressing critical environmental concerns. The integration of 3D design, mechanical work, IoT technologies, and artificial intelligence into a cohesive and functional robot showcases the multidisciplinary skills and creativity embedded in the project.

Your attention to detail in designing components such as the hand, thigh, shin, foot, shoulder, and backbone, coupled with the meticulous crafting using AutoCAD and Pepakura, demonstrates a level of craftsmanship that goes beyond technical proficiency.

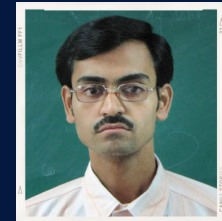
The use of Python for AI applications, including voice recognition, dynamic video display, and real-time voice conversation, adds a layer of sophistication to the project. Furthermore, the commitment to aesthetic appeal with the black and golden spray-painted finish and motion sensor-driven face movement sets the E3AN robot apart as an innovative and visually engaging creation.

As the project administrator, I commend each member of the team for your resilience, teamwork, and dedication to achieving the project's goals. Your success is not only a testament to your individual and collective abilities but also reflects positively on the academic ethos of our institution.

I encourage you to take pride in your accomplishments and view this project as a stepping stone to even greater endeavors. Your innovative approach to e-waste repurposing serves as an inspiration to your peers and future generations of students.

Once again, congratulations on a job well done! I look forward to seeing the impact your work will have on the broader technological and environmental landscape.

Best regards,

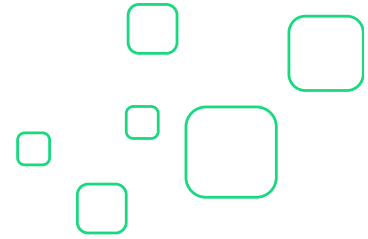


PROJECT SUPERVISOR

**MR SUJIT GOSWAMI
ASST. PROFESSOR
ECE DEPARTMENT**

PROJECT SUPERVISOR

ACKNOWLEDGEMENT



It is with immense gratitude and appreciation that we extend our heartfelt acknowledgments to those who have been instrumental in the successful completion of the E-Waste Robot Project, titled "E3AN: Transformative E-Waste Repurposing."

First and foremost, we express our deepest thanks to our project supervisor Mr. Sujit Goswami, for their unwavering support, guidance, and mentorship throughout the duration of this project. Their expertise, insightful feedback, and encouragement were pivotal in steering us through challenges and refining our approach.

We are indebted to the entire Electronics & Communication Engineering Department at Asansol Engineering College for providing us with a conducive academic environment and valuable resources that enabled us to undertake and complete this innovative project. The department's commitment to fostering a culture of creativity and exploration has been a driving force behind our endeavors.

Our sincere appreciation goes to Other faculty members for their valuable input and suggestions, which significantly enriched the project's conceptualization and execution. Their diverse perspectives and wealth of knowledge were integral to the project's success.

Special thanks are extended to our fellow students and peers who contributed to various aspects of the project, offering support, insights, and collaboration. The sense of camaraderie within the student community played a crucial role in overcoming challenges and celebrating achievements.

We express our gratitude to the technical support staff and the laboratory personnel of our college, whose assistance with equipment, tools, and facilities greatly facilitated the practical implementation of our project.

Finally, our heartfelt thanks go to our families and friends for their unwavering encouragement, understanding, and patience during the demanding phases of the project. Their support has been a source of strength and motivation.

This project has been a collective effort, and the successful completion would not have been possible without the collaborative spirit and encouragement from each of the individuals mentioned above.

Thank you for being an integral part of our journey.

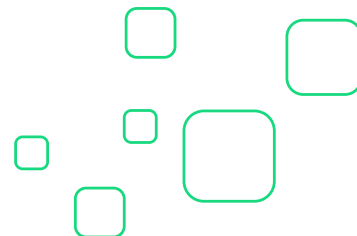
Sincerely,

Team E3AN

Department of Electronics & communication Engineering
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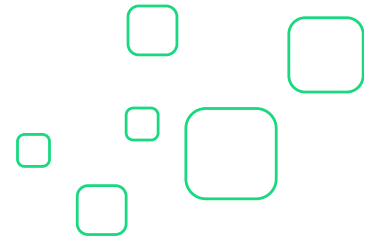


SESSION 2022-26

Signature

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ABSTRACT



Title: E3AN - Transformative E-Waste Repurposing

Project Administration:
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Department:
ELECTRONICS & COMMUNICATION ENGINEERING

Institution:
ASANSOL ENGINEERING COLLEGE

The E3AN project is a pioneering initiative aimed at repurposing non-functioning computers into a fully operational and aesthetically appealing robot. With a focus on sustainability and innovation, the project integrates various disciplines, including 3D design, mechanical work, Internet of Things (IoT) technologies, and artificial intelligence (AI).

The project commenced with meticulous 3D design using AutoCAD for components such as the hand, thigh, shin, foot, shoulder, and backbone. Pepakura paper crafting techniques were then employed to give life to the outer covering, ensuring precision and attention to detail. Mechanical work, involving welding, metal cutting, and shaping, transformed discarded electronic components into a cohesive and functional robotic entity.

The integration of IoT technologies, centered around Arduino as the central control unit, showcased the team's proficiency in harnessing electronic devices such as a motor driver, ultrasonic sensor, LED lights, and servo motor. Python, chosen for its versatility, was employed for AI applications, including voice recognition, dynamic video display, and real-time voice conversation.

The aesthetic appeal of the project was not overlooked, with the robot adorned in a striking black and golden spray-painted finish. Motion sensor-driven face movement added an interactive and engaging element, emphasizing the project's user-centric design.

This abstract encapsulates the multidisciplinary nature of the E3AN project, offering a glimpse into its technical intricacies, innovative solutions, and the team's commitment to sustainable technology. The project stands as a beacon of creativity, addressing the global challenge of e-waste through transformative repurposing.

Keywords: E-Waste, Robotics, Sustainability, 3D Design, IoT, Artificial Intelligence, Repurposing.



CONTENT

Certificate of Recommendation

Certificate of Approval

Acknowledgement

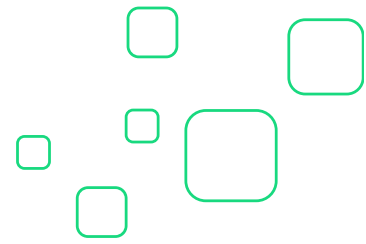
Abstract

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Introduction

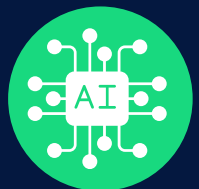


Introduction to Project E3AN: Revitalizing Old Computers into a Smart E-Waste Robot

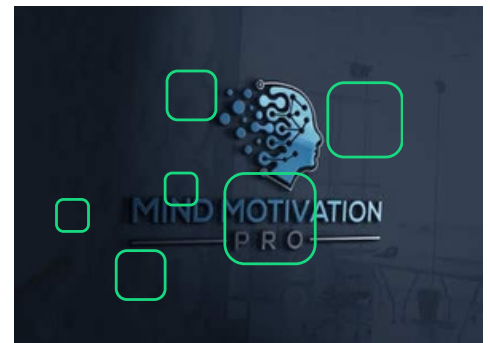
In an era of technological advancement, the imperative to address electronic waste (e-waste) has never been more crucial. Embracing this challenge, our team embarked on a groundbreaking project, breathing new life into non-functioning computers and transforming them into a state-of-the-art robotic marvel – Project E3AN.

- The journey began with salvaging old computers that had reached the end of their functional life. Undeterred by their initial state, we meticulously refurbished and repurposed these machines, fashioning a resilient core for our robot.
- The structural blueprint of E3AN took shape through intricate 3D designs crafted using AutoCAD, ensuring precision and efficiency. To embellish its exterior, we employed Pepakura, creating an aesthetically pleasing outer covering that complemented the robot's sleek functionality.
- For the internal framework, we opted for durability and lightness, employing an aluminum frame. The body of the robot was sheathed in aluminum sheets, providing both resilience and a modern, metallic finish.
- Integrating the power of the Internet of Things (IoT), we utilized Arduino and other electronic components to enable smart functionalities, pushing the boundaries of innovation. Java and Python programming languages played a pivotal role in imbuing E3AN with artificial intelligence and voice recognition capabilities, enhancing its adaptability and responsiveness.
- The video display, a crucial aspect of user interaction, was ingeniously realized by repurposing a computer monitor. This not only contributed to sustainability but also offered a familiar interface for users.
- To add the finishing touches, the robot was meticulously painted with a combination of black and golden spray paint, elevating its aesthetic appeal and ensuring it stood out as a testament to creativity and innovation.
- E3AN, the culmination of our endeavors, not only represents technological ingenuity but also boasts a dynamic face equipped with a motion sensor. This feature enables the robot's face to intuitively respond to the user's presence, creating a more immersive and engaging interaction.

In summary, Project E3AN stands as a testament to our commitment to sustainability, innovation, and the limitless possibilities that arise when creativity converges with technology.



Motivation of The Project



Project E3AN: Bridging Innovation and Sustainability

In the vast landscape of technological progress, there exists an undeniable responsibility to innovate with purpose, to craft solutions that not only push the boundaries of what is possible but also contribute to a sustainable future. It is within this ethos that Project E3AN was conceived – a testament to our unwavering commitment to redefine the possibilities of repurposing and reimagine the role of technology in addressing environmental challenges.

Motivated by a shared vision of a world where discarded electronic devices are not merely waste but potential sources of innovation, our team embarked on a transformative journey. The foundation of our motivation lies in the realization that the legacy of technology should not be confined to obsolescence but should instead be a continuum of ingenuity and adaptability.

As we resurrected non-functioning computers from the depths of e-waste, we found motivation in the prospect of breathing new life into discarded devices, transforming them into a tangible manifestation of our commitment to sustainability. The sheer challenge of revitalizing these machines fueled our determination to prove that innovation and eco-consciousness can coexist harmoniously.

The integration of advanced technologies such as IoT, artificial intelligence, and voice recognition became our driving force. In each line of code and every circuit connection, we found the motivation to push our creative boundaries, envisioning a robot that not only serves a purpose but does so with intelligence, responsiveness, and a touch of futuristic elegance.

The motivation behind Project E3AN extends beyond the functional aspects of the robot. With every meticulously designed 3D structure, each stroke of spray paint, and the careful consideration of every component, we aimed to inspire a paradigm shift in how we perceive and utilize technology. E3AN embodies the belief that innovation can be synonymous with sustainability, creating a ripple effect that challenges conventional norms.

The vision of Project E3AN is not merely confined to the confines of our workshop; it extends to a world where technology not only advances but does so responsibly. Our motivation is rooted in the conviction that E3AN can spark conversations, inspire like-minded individuals, and serve as a beacon for a future where innovation and environmental consciousness go hand in hand.

In essence, Project E3AN is more than a culmination of technical skills; it is a collective expression of our passion, resilience, and unwavering motivation to be architects of positive change. As we present our creation to the world, we invite you to join us in this journey, where motivation fuels innovation and sustainability becomes the cornerstone of progress.



LEARNING SKILLS

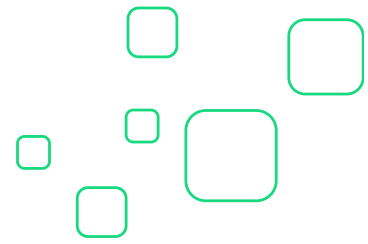


ACHEIVING HIGH GOAL



TEAM WORK

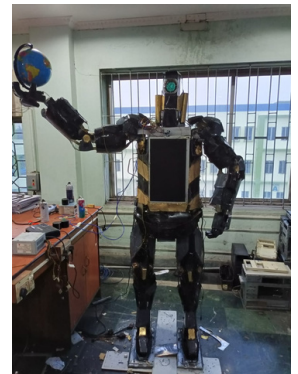
Basic Description of the Project



Project E3AN is a groundbreaking initiative that repurposes non-functioning computers, transforming them into a cutting-edge robotic marvel. The project encompasses a comprehensive approach to sustainability and innovation, from refurbishing old computers to utilizing advanced technologies such as 3D design, IoT applications, artificial intelligence, and voice recognition.

The structural foundation of E3AN involves meticulous 3D designs using AutoCAD, an aluminum frame for internal structure, and aluminum sheets for the robot's body. The outer covering is crafted with Pepakura, contributing to both functionality and aesthetic appeal.

Smart functionalities are achieved through the incorporation of Arduino and electronic components for IoT applications, while programming in Java and Python facilitates artificial intelligence and voice recognition capabilities. The robot repurposes a computer monitor for video display, contributing to both sustainability and user familiarity.

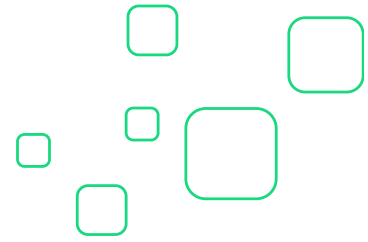


Description Here

E3AN is not only a technical achievement but also a creative endeavor. The robot is adorned with a sleek combination of black and golden spray paint, elevating its aesthetic appeal. Its dynamic face incorporates a motion sensor for responsive interactions, further enhancing the user experience.

In essence, Project E3AN stands as a symbol of our commitment to innovation and sustainability, demonstrating that discarded technology can be reimagined into a functional, intelligent, and aesthetically pleasing robotic solution.

3D Designing Internal core part



Introduction:

In the realm of technological innovation and sustainable practices, the significance of 3D design becomes particularly pronounced, especially in ambitious projects such as repurposing old computers into a multifunctional robotic solution. The integration of 3D design, particularly through tools like AutoCAD, plays a pivotal role in shaping the success and functionality of each component within the project. This introduction aims to elucidate the overarching importance of 3D design in the context of this e-waste project, emphasizing the compelling need for precision and intricate detailing in the design of essential components such as the hand, thigh, shin, foot, shoulder, and backbone.

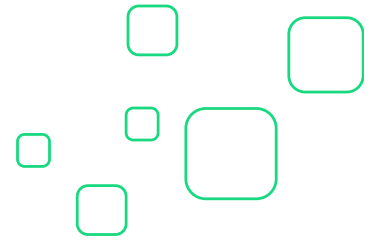


Significance of 3D Design:

The transition from two-dimensional conceptualization to three-dimensional modeling is fundamental in the realization of complex structures, offering a comprehensive perspective that transcends traditional design boundaries. In the context of repurposing old computers, the amalgamation of disparate components necessitates a meticulous approach that 3D design effortlessly facilitates. It allows for a more holistic visualization of the end product, offering engineers and designers the ability to foresee challenges, identify potential interferences, and optimize the overall structure before implementation.

Moreover, 3D design serves as a communication tool, providing a clear and tangible representation of the envisioned components. This aids collaboration among team members, stakeholders, and even potential users, fostering a shared understanding of the project's intricacies.

3D Designing Internal core part



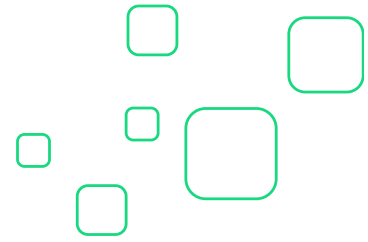
Precision and Detail in Component Design:

The intricacy of designing components such as the hand, thigh, shin, foot, shoulder, and backbone demands a level of precision that is paramount to the project's success. Each component, though unique in its functionality, must seamlessly integrate with others to ensure the robot's overall efficacy.

1. **Hand:** Precision is crucial to ensure the hand's dexterity, allowing for intricate movements and grasping capabilities.
2. **Thigh and Shin:** These components require careful design to achieve a balance between structural integrity and mobility, especially considering the dynamic nature of the robot's movements.
3. **Foot:** The foot, serving as the point of contact with the ground, necessitates detailed design to guarantee stability, weight distribution, and adaptability to various surfaces.
4. **Shoulder and Backbone:** These components play a central role in supporting the overall structure, requiring precision to maintain balance, and enabling fluid movement.

In essence, the precision and detailing in the design are not merely aesthetic considerations but fundamental to the functionality and performance of the robotic system. The success of the e-waste project hinges on the seamless integration of these components, and 3D design, particularly through tools like AutoCAD, emerges as the linchpin in achieving this delicate balance between innovation and precision.

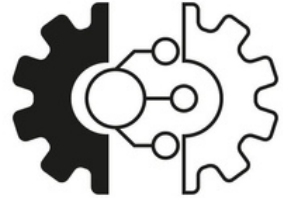




Purpose of AutoCad

AutoCAD was chosen as the primary tool for 3D design in the e-waste project due to its unparalleled capabilities, versatility, and alignment with the specific objectives of the project. The decision to utilize AutoCAD was rooted in its applicability to the intricacies of repurposing old computers into a sophisticated robotic solution. Here are the key reasons why AutoCAD was deemed the most fitting tool for this endeavor:

1. **Industry Standard for Precision:** AutoCAD is an industry-standard software renowned for its precision and accuracy in design. Given the need for detailed schematics in the creation of hand, thigh, shin, foot, shoulder, and backbone components, AutoCAD's reputation for maintaining high levels of accuracy was instrumental. This precision is crucial for ensuring that each component fits seamlessly with others and contributes to the overall functionality of the robotic system.
2. **Comprehensive 3D Modeling Capabilities:** AutoCAD's robust 3D modeling capabilities were a decisive factor in its selection. The e-waste project demanded a tool that could translate intricate 2D designs into comprehensive 3D models. AutoCAD's ability to create, modify, and visualize complex three-dimensional structures allowed the project team to iterate through various design options and refine the components with a high degree of detail.
3. **Ease of Collaboration and Documentation:** AutoCAD's collaborative features and comprehensive documentation tools were critical for effective communication within the project team. The software facilitates the creation of detailed documentation, ensuring that design intent is conveyed clearly. This is especially vital in a project with multiple components, where precise communication is paramount to achieving the desired end result.
4. **Adaptability for Varied Component Designs:** The diverse nature of the components, ranging from the dynamic movements of the hand to the structural considerations of the backbone, required a tool that could adapt to various design challenges. AutoCAD's flexibility in handling different design requirements, whether organic or geometric, made it the ideal choice for addressing the varied needs of the e-waste project.
5. **Integration with Other Design Tools:** AutoCAD seamlessly integrates with other design and engineering tools, fostering a collaborative environment. This interoperability ensures that the 3D designs created in AutoCAD can be easily transferred and utilized in conjunction with other tools employed in the project, contributing to a more cohesive and efficient design process.



Integration and Iterations in AutoCAD

Integrating individual designs for the hand, thigh, shin, foot, shoulder, and backbone into a cohesive structure was a critical aspect of ensuring the success and functionality of our e-waste project. AutoCAD played a pivotal role in this integration process, providing the tools and features necessary to achieve a seamless fit among the diverse components. Here's how the integration was facilitated:

- Unified 3D Model:

AutoCAD allowed for the creation of a unified 3D model that incorporated all individual components. The software's ability to manage complex geometries and assemble various parts into a single model ensured a holistic representation of the entire robotic structure.

- Component Alignment:

AutoCAD's precision and measurement tools were utilized to align and position each component accurately within the 3D space. This ensured that the hand, thigh, shin, foot, shoulder, and backbone components fit together seamlessly, both visually and functionally.

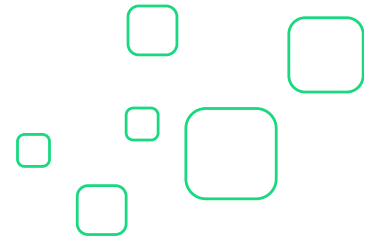
- Interference Detection:

AutoCAD's interference detection capabilities were crucial in identifying potential clashes or conflicts between components. This feature allowed the project team to visualize and rectify any overlaps or interferences, ensuring that the integrated design would not face physical constraints or conflicts during fabrication.

- Parametric Design:

The parametric design capabilities of AutoCAD enabled the establishment of relationships between different components. This meant that modifications to one part of the design automatically propagated to other related elements. This feature was instrumental in maintaining consistency and coherence throughout the entire structure.

Outer Cover Design: Pepakura



Introduction to Pepakura in Outer Cover Design:

In the intricate realm of design and innovation, the importance of outer cover aesthetics cannot be overstated. As part of the endeavor to repurpose old computers into a functional and visually striking robotic solution, the incorporation of Pepakura emerges as a transformative tool in the crafting of the outer cover for the entire body. Pepakura, a versatile paper-crafting software, not only stands as a testament to creativity but also plays a pivotal role in achieving the delicate balance between visual appeal and structural integrity in the e-waste project.



Versatility of Pepakura

Pepakura, derived from the Japanese words "pepa" (paper) and "kura" (crafting), is renowned for its versatility in transforming digital 3D models into paper patterns that can be folded and assembled into tangible objects. Originally conceived for hobbyists creating intricate paper models, Pepakura's adaptability transcends its traditional usage, finding innovative applications in various design domains.

Relevance to Outer Cover Design:

The outer covering of the robotic structure is not merely a protective layer but a canvas for creativity and a critical element in the project's success. Pepakura becomes particularly relevant in this context due to several key factors:

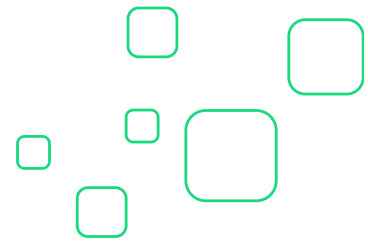
- **Visual Appeal**
- **Structural Soundness**
- **Adaptability to Recycled Materials**
- **User-Friendly Iterations**

Significance of Outer Cover Design:

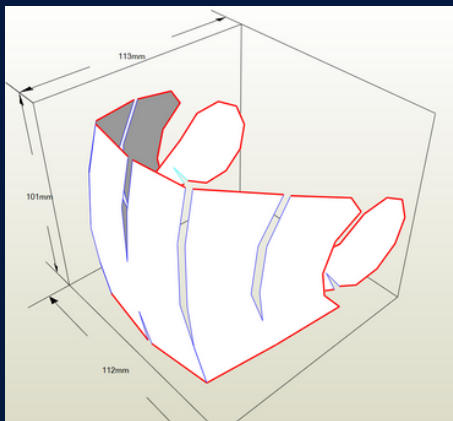
The outer covering is the first visual encounter users have with the robotic solution. Beyond aesthetics, it serves as a protective layer, shielding internal components from external elements. The integration of Pepakura not only addresses these functional aspects but elevates the outer cover to a work of art, contributing to the overall user experience and the project's environmental narrative.

In essence, Pepakura's role in the e-waste project extends beyond mere paper crafting; it becomes a catalyst for transforming recycled materials into a visually stunning and structurally sound outer covering. Its versatility and compatibility with sustainability objectives position Pepakura as an invaluable tool in the pursuit of innovation, merging the realms of artistry and engineering in the repurposing of old computers into a sophisticated and visually appealing robotic solution.

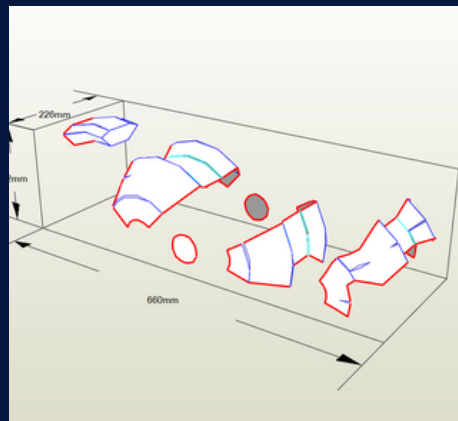
Outer Cover Design Using PEPAKURA



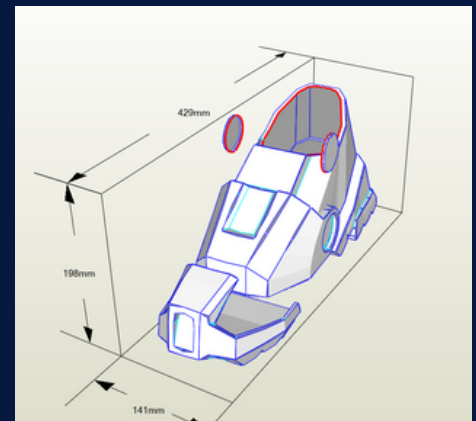
ELBOW



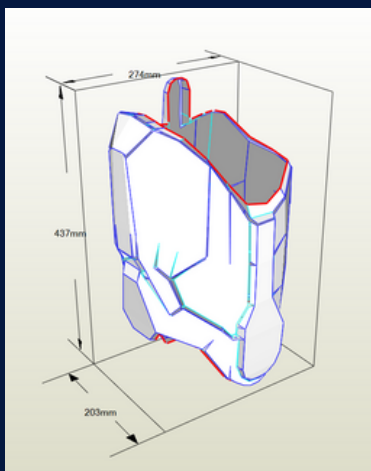
SHOULDER



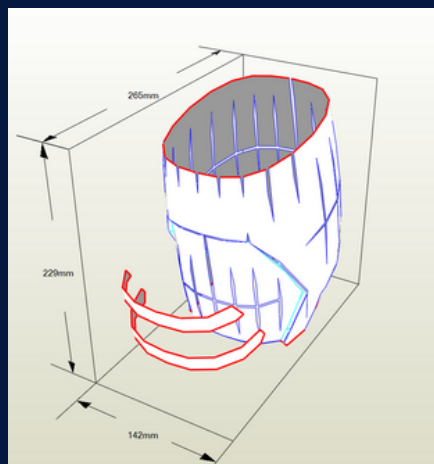
BOOT



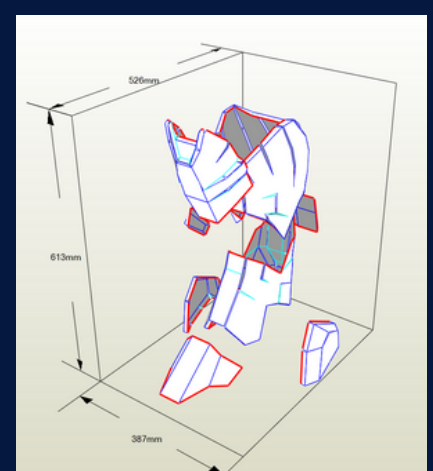
THIGH



BICEP



SHIN



Integration and Iterations in Pepakura



The integration of individual outer cover designs into a cohesive whole within the e-waste robot project was a complex yet rewarding process, facilitated by Pepakura. The software's ability to convert 3D digital models into precise paper patterns allowed for the systematic assembly of diverse components. However, the journey from individual designs to a unified outer cover did present specific challenges, all of which were met and overcome through strategic design considerations and Pepakura's features.

1. Component Synchronization:

Each component, whether it be the hand, thigh, shin, foot, shoulder, or backbone, had its unique design intricacies. Pepakura facilitated the integration process by providing a platform to synchronize these designs. However, aligning different components to ensure a seamless fit required meticulous attention to detail. Adjustments in Pepakura allowed for fine-tuning, ensuring that the outer covers aligned precisely with one another to create a harmonious and unified aesthetic.

- 2. Scaling and Proportional Consistency: Achieving uniformity in size and proportions across different components was a significant challenge. Pepakura's scaling features became instrumental in maintaining consistency. Careful measurements and scaling adjustments ensured that the outer covers maintained proportional harmony, even as they covered components with different shapes and functionalities.

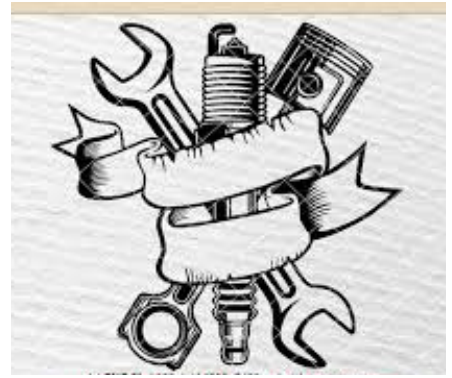
3. Complex Geometries and Folding Sequences:

The e-waste robot's intricate design posed a challenge in terms of the complex geometries of individual components and the corresponding folding sequences needed for assembly. Pepakura's user-friendly interface allowed for the creation of intricate folding patterns, ensuring that each outer cover's assembly followed a logical and manageable sequence. This was crucial in streamlining the fabrication process and ensuring that the outer covers adhered to the intended form.

4. Iterative Refinement:

As with any design process, iterations were a fundamental part of achieving the desired outcome. Pepakura's iterative refinement capabilities allowed the team to make adjustments to individual components and observe real-time changes in the assembled model. This iterative approach was vital in addressing any misalignments or inconsistencies that arose during the integration process.

Introduction to Mechanical Work in E-Waste Robot Project



Introduction

In the realm of innovative engineering projects, the pivotal role of mechanical work stands as the linchpin in translating conceptual designs into tangible and functional realities. The completion of your e-waste robot project is a testament to the instrumental contribution of mechanical work, where the utilization of an array of tools, including welding machines, metal cutters, hack-saws, hammers, and more, was indispensable. This introduction aims to underscore the necessity and significance of employing various mechanical tools in transforming initial conceptual designs into the robust and functional components that constitute the e-waste robot.

The Essence of Mechanical Work

Mechanical work in the context of the e-waste robot project encompasses a spectrum of processes, techniques, and tools that collectively breathe life into the conceptualized designs. Beyond the digital realm of design software and theoretical blueprints, it is the mechanical work that bestows substance and functionality upon the envisioned robotic solution.

Necessity of Mechanical Tools:

The utilization of various mechanical tools is necessitated by the intricate nature of the e-waste robot project, which involves repurposing old computers into a sophisticated, functional, and aesthetically pleasing robotic solution. Each tool, from welding machines to metal cutters, hack-saws, and hammers, serves a distinct purpose in the transformation process.

- **Welding Machines for Structural Integrity**
- **Metal Cutters and Hack-Saws for Precision**
- **Hammers for Assembly and Refinement**

Utilization of Metal Cutters and Hack-Saws in Shaping and Refining Metal Components for the E3AN:



The use of metal cutters and hack-saws in your e-waste robot project played a pivotal role in shaping and refining metal components, contributing to the precision required for creating individual parts such as limbs, frames, and connectors. This section details how these tools were employed to achieve accuracy and finesse in the fabrication process.

1. Material Selection and Preparation: Before diving into the cutting process, meticulous consideration was given to the selection of recycled materials from old computers. Metal cutters and hack-saws were chosen based on the type and thickness of the materials, ensuring that the tools were well-suited for the specific characteristics of the recycled metal components.

2. Cutting Limbs and Appendages: Metal cutters were employed to cut precise shapes for limbs and appendages, ensuring uniformity and consistency in the robotic structure. The ability to control the cutting process allowed for the creation of identical parts, contributing to the symmetrical design of the robot. The hack-saw, with its manual control, was utilized for more intricate cuts requiring detailed and careful execution.

3. Frame Fabrication: The construction of the robot's frame required precise cutting of metal components to achieve the desired dimensions. Metal cutters were utilized to make straight and angular cuts for frame elements, ensuring that each piece fit together seamlessly during assembly. The hack-saw, with its versatility in cutting various profiles, provided a complementary tool for refining edges and achieving the necessary precision.

4. Refining Edges and Surfaces: The hack-saw, known for its ability to create smoother finishes, was used for refining edges and surfaces of cut metal components. This step was crucial for eliminating sharp edges, ensuring user safety, and enhancing the overall aesthetics of the robot. Metal cutters, with their cutting precision, facilitated the initial shaping of components, while the hack-saw added the finishing touch.

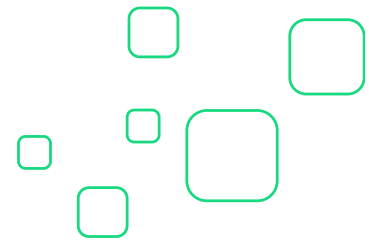
Role of Hammers and Shaping Tools in Refinement of the Robot's Appearance:



The refinement of your e-waste robot's appearance was a meticulous process that involved the strategic use of hammers and other shaping tools. These tools played a crucial role in detailing, smoothing rough edges, and ensuring a polished finish, elevating the overall aesthetics of the repurposed components.

1. **Detailing and Shaping of Components:** Hammers, along with other shaping tools, were employed to add intricate details to various components of the robot. Detailing played a significant role in enhancing the visual appeal, giving a distinct character to different parts such as limbs, frames, and connectors. This detailing process involved controlled hammer strikes and shaping techniques to create patterns, textures, or contours that contributed to the robot's overall design.
2. **Smoothing Rough Edges:** During the fabrication process, especially when using recycled materials from old computers, rough edges and irregularities are inevitable. Hammers were utilized to smooth out these rough edges systematically. Shaping tools, which could include files or grinders, complemented the hammering process by refining the surfaces, ensuring a uniform and polished appearance.
3. **Forming Curves and Contours:** Shaping tools, guided by skilled hands and creative vision, were instrumental in forming curves and contours within the robot's design. Hammers, in particular, were employed to shape metal components, ensuring that they seamlessly fit together while adding an artistic dimension to the robot's overall aesthetic.
4. **Repairing and Modifying Components:** In instances where components needed adjustments or repairs, hammers and shaping tools were indispensable. The ability to modify and reshape components during the refinement process allowed for continuous improvements. This iterative approach ensured that the robot's appearance met evolving design standards and addressed any unforeseen challenges.

Utilization of Various Mechanical Tools in the Assembly Process for the E-Waste Robot Project:



The assembly process of your e-waste robot project involved the strategic use of various mechanical tools to join different components together. Techniques such as riveting, bolting, and hammering played a crucial role in creating secure and reliable joints, ensuring the overall functionality and structural integrity of the robot.

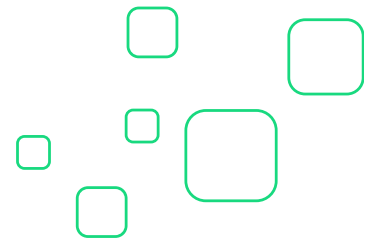
1. Riveting for Secure Joints: Riveting was employed as a technique to create secure and permanent joints in the assembly of the e-waste robot. This process involved using a rivet gun to insert rivets into pre-drilled holes in the components being joined. The rivets were then expanded, creating a tight and sturdy connection. This method was particularly effective in areas where welding might not be suitable or for components that required detachability for maintenance or upgrades.

2. Bolting for Versatility and Flexibility: Bolting provided versatility and flexibility in the assembly process. It involved using nuts and bolts to secure components together. This technique allowed for disassembly and reassembly, making it useful for adjustments, repairs, or modifications. Bolting was employed in areas where a balance between strength and flexibility was required, and it facilitated the creation of robust joints that could withstand various mechanical stresses.

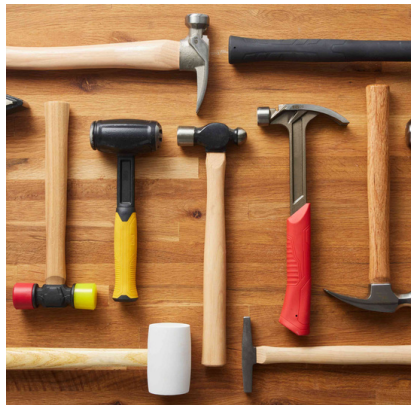
3. Integration of Different Techniques: The assembly process was not limited to a single technique but rather involved the integration of different methods based on the requirements of specific components. For instance, areas with high stress and load demands might involve welding, while detachable or adjustable parts utilized riveting or bolting. This strategic combination of techniques allowed for a tailored approach to assembly.

4. Importance of Secure Joints: Secure joints were paramount for the robot's overall functionality. The reliability of the joints determined the structural stability of the robot, directly impacting its ability to move, perform tasks, and withstand operational stresses. The choice of mechanical tools and techniques was guided by the need to create joints that could endure the dynamic requirements of the robot's functions, ensuring its safe and efficient operation.

Mechanicle Equipments Used



HAMMER



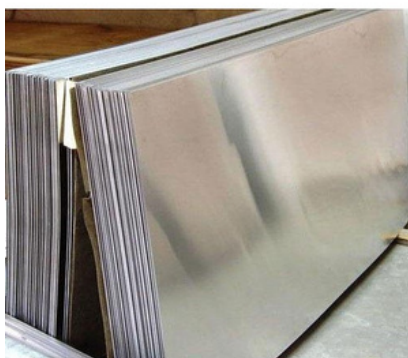
CUTTER



GRINDER



SHEET



DRILL MACHINE



SCREW DRIVER



Introduction: Enhancing E-Waste Robot with IoT Technologies



In the pursuit of transforming discarded electronics into a cutting-edge technological marvel, the integration of various IoT electronic devices stands as a cornerstone in the development of our e-waste robot project. The infusion of smart technologies not only elevates the functionality of the robot but also enhances its interactivity, turning a collection of repurposed components into a sophisticated and responsive robotic solution.

The Significance of IoT Integration:

In the landscape of modern robotics, the integration of IoT devices marks a paradigm shift, breathing life into the concept of repurposing old computers. The adoption of smart technologies, such as Arduino for centralized control, motor drivers for precise movements, ultra sensors for environmental awareness, LED lights for expressive visuals, and servo motors for dynamic actions, signifies a strategic leap forward. This amalgamation of diverse IoT elements adds layers of functionality, responsiveness, and aesthetic appeal to the e-waste robot, transforming it into a dynamic and interactive entity.

Enhancing Functionality through Centralized Control:

At the heart of this transformation lies the utilization of Arduino as the central nervous system. This microcontroller platform serves as the orchestrator, seamlessly integrating and controlling various components. The significance of Arduino extends beyond mere control; it empowers the robot with the ability to execute complex tasks, respond to environmental stimuli, and adapt its actions based on real-time data. The centralized control offered by Arduino forms the foundation for a highly functional and versatile robotic system.

Precision and Motion with Motor Drivers:

The integration of motor drivers introduces a level of precision and control to the e-waste robot's movements. Specifically, in the rotation of the robot's head, the motor driver acts as the catalyst for fluid and calculated motion. This not only adds a dynamic dimension to the robot's interactions but also showcases the potential for repurposed electronics to perform intricate and intentional actions. The inclusion of motor drivers demonstrates the transformative power of IoT technologies in turning salvaged materials into responsive and purposeful components.

Introduction: Enhancing E-Waste Robot with IoT Technologies



Environmental Awareness with Ultra Sensors:

IoT's role in enhancing interactivity is particularly evident through the integration of ultra sensors. These sensors imbue the e-waste robot with a heightened environmental awareness, allowing it to detect and respond to the presence and movements of users. The deployment of ultra sensors serves as a testament to the adaptability of smart technologies, enabling the robot to engage with its surroundings and users in a dynamic and responsive manner.

Expressive Visuals with LED Lights:

The incorporation of LED lights introduces an element of expressive visuals to the e-waste robot. Beyond mere functionality, the robot becomes visually engaging, communicating its status and responses through dynamic lighting effects. The synergy between smart control systems and LED lights showcases the potential for repurposed electronics not just to perform tasks but to convey a sense of personality and interaction, making the robot more relatable and engaging.

Dynamic Actions with Servo Motors:

The inclusion of servo motors takes the e-waste robot's actions to a new level of dynamism. Specifically, the rotation of the globe demonstrates how IoT technologies can create visually striking and interactive features. The servo motor, controlled by the central Arduino unit, enables the robot to execute controlled and purposeful movements, adding an aesthetic and engaging aspect to its overall design.

Conclusion: A Technological Renaissance from E-Waste:

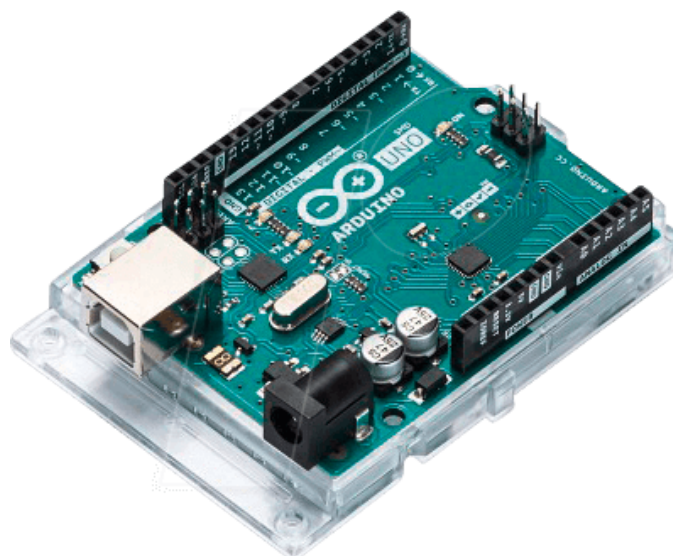
In conclusion, the integration of different IoT electronic devices in your e-waste robot project represents a technological renaissance, breathing new life into discarded electronics. The significance of incorporating smart technologies lies not only in the enhanced functionality and precision of the robot but also in its newfound interactivity. Through the strategic use of Arduino, motor drivers, ultra sensors, LED lights, and servo motors, the project showcases the transformative power of IoT, proving that repurposed materials can become the foundation for sophisticated and responsive robotic solutions. As the e-waste robot stands at the intersection of innovation and sustainability, it serves as a testament to the potential within discarded electronics and the possibilities that emerge when smart technologies are harnessed for creative and purposeful endeavors.

The Central Role of Arduino: The Brain of the E-Waste Robot

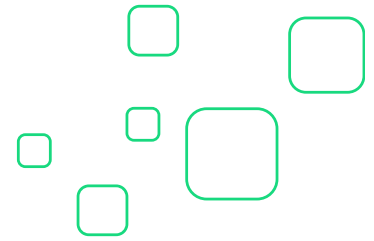


In the realm of your e-waste robot project, Arduino assumes a central role, acting as the veritable brain that orchestrates and harmonizes the multifaceted functions of diverse components. As the central processing unit, Arduino not only imparts intelligence to the robot but also serves as the linchpin ensuring synchronized and cohesive operation of the entire system.

1. **Centralized Control Hub:** At its core, Arduino functions as a centralized control hub, providing a singular platform for programming and managing the interactions of various electronic components. This centralized nature is fundamental in streamlining the project's control architecture, offering a unified environment where different devices can communicate, collaborate, and collectively contribute to the robot's functionality.
2. **Programming Flexibility and Customization:** Arduino's versatility and ease of programming play a pivotal role in shaping the behavior of the e-waste robot. The open-source nature of Arduino facilitates custom code development, enabling tailored solutions for each component. This flexibility ensures that the robot's actions are not static but can be adapted, refined, and expanded over time to meet evolving project requirements.
3. **Coordination of Motorized Movements:** In the context of the e-waste robot, Arduino becomes the maestro orchestrating motorized movements with precision. For example, in controlling the rotation of the robot's head, Arduino coordinates with motor drivers, dictating the speed, direction, and duration of movement. This coordination ensures that the head rotates seamlessly, adding a dynamic and purposeful dimension to the robot's interactions.



Implementation of Motor Driver for Head Rotation in the E-Waste Robot

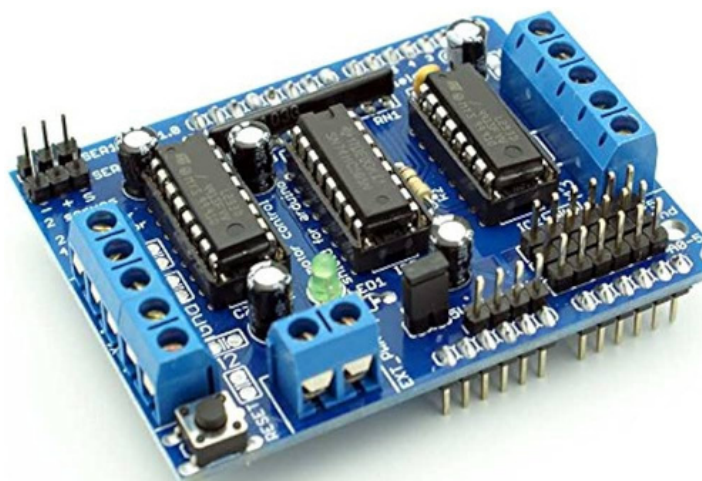


1. Head Rotation Mechanism: The implementation of a motor driver to facilitate the rotation of the robot's head is a crucial element in bringing dynamic movement to the e-waste robot. The head rotation mechanism adds an interactive dimension to the robot's capabilities, allowing it to turn and engage with its surroundings. This section delves into the key aspects of this implementation, discussing both the choice of motor driver and its integration into the overall system.

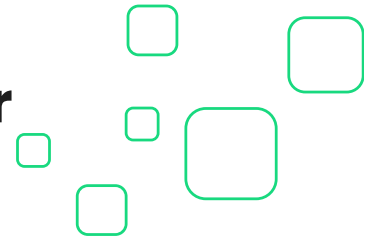
2. Choice of Motor Driver: The selection of an appropriate motor driver is pivotal to ensuring the success of the head rotation mechanism. Several factors were considered in making this choice:

- **2.1 Compatibility with Head Mechanism:** The motor driver was chosen based on its compatibility with the specific requirements of the head rotation mechanism. This involved assessing the type of motor used for the head movement, its voltage and current requirements, and the overall mechanical design. The goal was to ensure seamless integration without compromising the structural integrity of the robot.

2.2 Desired Speed and Precision: The desired speed and precision of the head rotation played a crucial role in selecting the motor driver. For smooth and controlled movements, a motor driver capable of providing variable speed control and precise position feedback was preferred. This ensured that the head could rotate at an optimal speed, aligning with the intended user interactions and overall aesthetic of the robot.



Implementation of Motor Driver for Head Rotation in the E-Waste Robot



3. Integration with Arduino: Once the motor driver was selected based on these criteria, the next step was the seamless integration with Arduino:

- **3.1 Wiring and Connections:** The wiring and connections between Arduino, the motor driver, and the motor itself were established with meticulous attention to detail. This involved connecting the appropriate pins, ensuring the correct polarity, and integrating any necessary power supply for both the Arduino and the motor driver.
- **3.2 Programming Logic:** Arduino's programming logic was developed to control the motor driver effectively. The code included commands for starting, stopping, and controlling the speed and direction of the motor. Special attention was given to programming routines that enabled smooth acceleration and deceleration, contributing to the desired precision in head rotation.

4. Testing and Calibration: Rigorous testing and calibration procedures were implemented to ensure the seamless functioning of the head rotation mechanism. This involved iterative testing, adjustment of parameters in the code, and fine-tuning of the motor driver settings to achieve the desired speed, precision, and reliability in head movements.

5. Maintenance Considerations: Considerations for the long-term maintenance and robustness of the head rotation mechanism were incorporated into the design. This included implementing safety features, such as current limiting, and providing clear instructions for any necessary maintenance or troubleshooting procedures.

Conclusion: Dynamic Head Rotation as a Key Feature:

In conclusion, the implementation of a motor driver for the head rotation mechanism in the e-waste robot project showcases a meticulous approach to both component selection and integration. The choice of a motor driver based on compatibility, speed, and precision considerations, coupled with the seamless integration with Arduino, demonstrates a commitment to creating a dynamic and responsive head movement feature. As a key element of user interaction and the overall aesthetic appeal of the robot, the head rotation mechanism exemplifies the transformative power of repurposed electronics when coupled with intelligent control systems.

Integration of LED Lights for Facial Illumination in the E-Waste Robot



1. Aesthetic Enhancement through LED Lights:

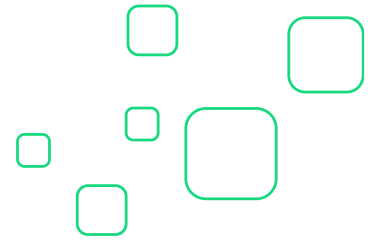
The integration of LED lights for facial illumination in your e-waste robot project adds a compelling aesthetic dimension, enhancing the visual appeal of the robot and providing a dynamic means of expression. This section explores the considerations and decisions involved in selecting and incorporating LED lights into the facial design.

2. Considerations in LED Lights Selection:

- 2.1 Color Palette: One of the first considerations in the selection of LED lights for facial illumination is the color palette. The choice of colors contributes significantly to the robot's personality and the ambiance it creates. Considerations for color psychology and the intended emotional impact on users guided the selection of LED colors. For example, warm colors like red and orange might evoke a friendly and inviting feel, while cool colors like blue and green might convey a more futuristic or calming atmosphere.
- 2.2 Brightness Levels: The brightness levels of the selected LED lights were carefully considered to achieve the desired visual impact without causing discomfort. Striking the right balance ensured that the facial illumination was vibrant and noticeable without being overpowering. This consideration is crucial, especially when the robot interacts with users in various lighting conditions.
- 2.3 Energy Efficiency: Given the sustainable and eco-friendly focus of the e-waste robot project, energy efficiency was a paramount consideration in LED light selection. Opting for energy-efficient LEDs not only aligns with the project's environmental ethos but also contributes to prolonged operation without a significant increase in power consumption. This consideration aligns with the overarching goal of transforming discarded electronics into a sustainable and energy-conscious robotic solution.



Integration of LED Lights for Facial Illumination in the E-Waste Robot



3. Dynamic Lighting Effects:

- 3.1 Synchronization with Movements: The LED lights were programmed to synchronize with the robot's movements, creating a cohesive and dynamic visual experience. For instance, the lights could change color or intensity in tandem with the rotation of the robot's head or respond to specific user interactions. This synchronization not only added an artistic flair to the robot's expressions but also showcased the interactive capabilities of the integrated LED system.
- 3.2 Customizable Patterns: The programmability of LED lights allowed for the creation of customizable lighting patterns. Arduino's control logic enabled the execution of intricate patterns or sequences, adding versatility to the robot's visual expressions. This dynamic aspect contributed to the overall engagement and appeal of the e-waste robot.

4. Testing and Iterative Refinement:

- 4.1 Iterative Testing: Rigorous testing of the LED lights was conducted in various scenarios to assess their performance and visual impact. Iterative testing allowed for adjustments in brightness, color transitions, and synchronization with other components, ensuring that the facial illumination met the desired aesthetic standards.
- 4.2 User Feedback Integration: User feedback played a vital role in refining the LED lighting system. Insights from users regarding the visual impact, color preferences, and overall experience with the illuminated facial features informed further adjustments and refinements. This user-centric approach ensured that the aesthetic appeal of the robot resonated with its audience.

5. Conclusion: Aesthetic Brilliance from Repurposed Electronics:

In conclusion, the integration of LED lights for facial illumination in the e-waste robot project exemplifies a thoughtful and creative approach to repurposing discarded electronics. The considerations in LED selection for color, brightness, and energy efficiency, coupled with the dynamic lighting effects and synchronization with movements, contribute to the aesthetic brilliance of the robot. As an essential element of user engagement, the LED lights showcase the transformative power of sustainable technology, turning e-waste into an expressive and visually striking robotic solution.

Implementation of Servo Motor for Globe Rotation in the E-Waste Robot



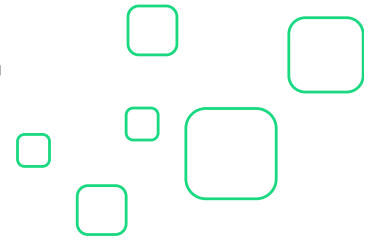
1. Dynamic Element: The incorporation of a servo motor to rotate the globe in your e-waste robot project introduces a dynamic and captivating element, elevating the robot's overall appeal and interaction capabilities. This section delves into the implementation details, focusing on the choice of a servo motor for its precise and controlled rotational movement.

2. Choice of Servo Motor:

- **2.1 Precise Rotational Movement:** The choice of a servo motor was driven by the need for precise and controlled rotational movement. Servo motors are known for their accuracy in positioning and controlled rotation, making them an ideal choice for applications where specific angles or positions need to be achieved. In the context of the e-waste robot project, this precision was crucial to executing smooth and purposeful rotations of the globe.
- **2.2 Compact Design:** Servo motors are inherently compact, making them suitable for applications with limited space, such as the rotating globe feature. The compact design ensures that the servo motor can be seamlessly integrated into the robot's structure without compromising the overall aesthetic or functionality.
- **2.3 Compatibility with Arduino Control:** Servo motors are compatible with Arduino and other microcontrollers, making them an ideal choice for projects where centralized control is essential. The seamless integration with Arduino allowed for programmable control over the servo motor, enabling dynamic and synchronized movements in response to user interactions or predefined sequences.



Implementation of Servo Motor for Globe Rotation in the E-Waste Robot



3. Integration with Arduino:

- 3.1 Wiring and Connections: The servo motor was integrated into the e-waste robot's framework through meticulous wiring and connections. The necessary power and control lines were connected to Arduino, ensuring a reliable and stable electrical connection. Attention to detail in wiring prevented signal interference and ensured that the servo motor responded accurately to control signals.
- 3.2 Programming Logic: Arduino's programming logic was developed to control the servo motor effectively. The code included instructions for starting, stopping, and controlling the rotational angle of the servo motor. The precise positioning capabilities of the servo motor allowed for specific angles to be achieved, enhancing the visual impact of the rotating globe.

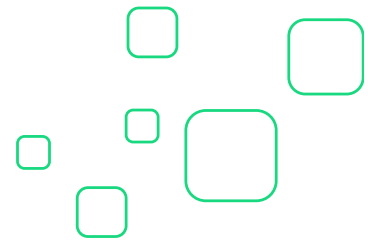
4. Controlled and Purposeful Movements:

- 4.1 Angular Range and Speed Control: The servo motor's angular range and speed were carefully calibrated to achieve controlled and purposeful movements. This involved programming the servo motor to rotate within specific limits and controlling the speed to ensure that the globe rotation was both visually engaging and synchronized with other elements of the robot.
- 4.2 Synchronization with Other Components: An essential aspect of the implementation was the synchronization of the servo motor with other components, such as LED lights or head rotation. This synchronization created a cohesive and dynamic visual experience, enhancing the overall interactivity and appeal of the e-waste robot.

6. Conclusion: Dynamic Elegance from Repurposed Components:

In conclusion, the implementation of a servo motor for the rotation of the globe in your e-waste robot project exemplifies a strategic and purposeful use of repurposed components. The choice of a servo motor for its precise and controlled rotational movement, coupled with meticulous integration with Arduino, contributes to the dynamic elegance of the robot. As a key visual element, the rotating globe enhances user engagement and showcases the transformative power of sustainable technology, turning discarded electronics into a captivating and interactive robotic solution.

Harnessing AI with Python: Transforming the E-Waste Robot Project



Introduction:

The integration of Python into our e-waste robot project represents a pivotal step forward in unlocking the potential of artificial intelligence (AI) to control the entire system. Python's versatility, coupled with its robust AI libraries, has played a foundational role in bringing to life key functionalities such as voice recognition, dynamic video display, and real-time voice conversation. This report provides a comprehensive exploration of how Python serves as the backbone, infusing intelligence and interactivity into the e-waste robot.

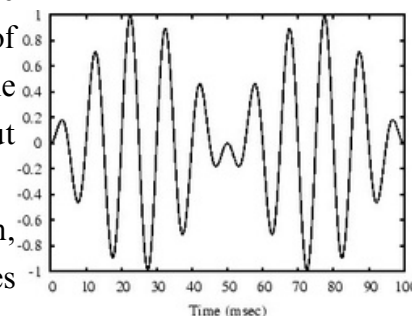


1. Python in Voice Recognition:

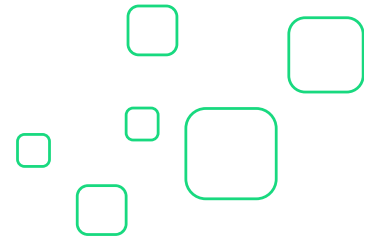
1. Python in Voice Recognition:

The implementation of Python for voice recognition marks a significant stride in creating an intuitive and responsive human-machine interface. Leveraging Python's rich ecosystem of speech recognition libraries, the robot can interpret and act upon spoken commands. The adaptability of Python allows for the seamless integration of these libraries into the overall system, enabling the robot to recognize and respond to user input effectively.

Python facilitates the training of AI models for voice recognition, allowing for iterative refinement and adaptation. This capability ensures that the robot becomes attuned to the nuances of user commands, contributing to a more natural and engaging interaction. The flexibility of Python in handling voice inputs positions the e-waste robot as an accessible and user-friendly device.



Harnessing AI with Python: Transforming the E-Waste Robot Project



2. Python for Dynamic Video Display:

Python's prowess in computer vision, exemplified through libraries like OpenCV, empowers the e-waste robot with dynamic video display capabilities. The integration of Python with computer vision libraries allows the robot to interpret visual inputs, making it responsive to its surroundings. This functionality is crucial for applications such as object recognition, enhancing the robot's ability to interact meaningfully with its environment.

Moreover, Python's scripting capabilities enable dynamic video rendering, facilitating real-time adjustments to video parameters. This responsiveness enhances the user experience, creating a visually engaging interface. The versatility of Python in manipulating video displays positions the e-waste robot as a technologically advanced and interactive system.



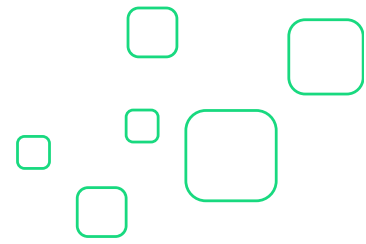
3. Real-Time Voice Conversation with Python:

Python's role extends to enabling real-time voice conversations, adding a layer of interactivity and natural communication to the e-waste robot. Leveraging Python's libraries for text-to-speech (TTS) and speech-to-text (STT), the robot can engage in bidirectional communication with users. The built-in feedback mechanism allows Python to process spoken words into textual information and vice versa, facilitating coherent and dynamic conversations.

The integration of natural language processing (NLP) techniques in Python further enhances the contextual understanding of the e-waste robot during voice conversations. This elevates the quality of interactions, making the robot more adept at interpreting user intent and responding contextually. Python's capabilities in NLP contribute to the creation of a more human-like and intelligent communication system.



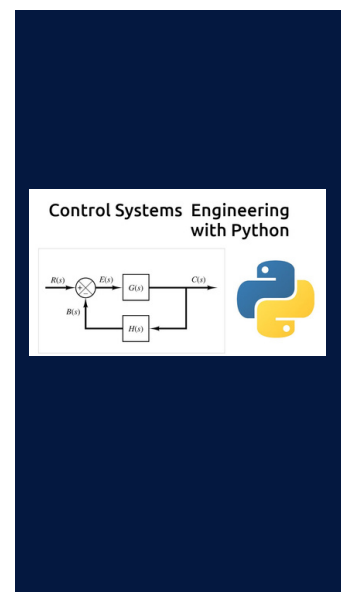
Harnessing AI with Python: Transforming the E-Waste Robot Project



4. Python Scripting for System Control:

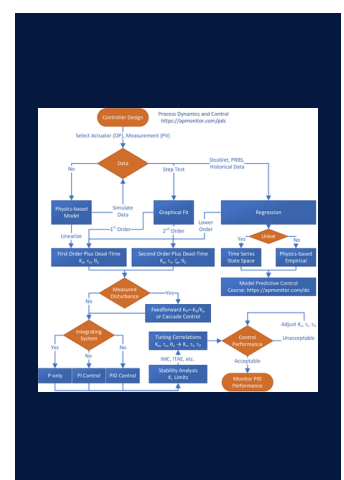
Python emerges as the central control mechanism for the entire e-waste robot system through its scripting capabilities. Python scripts orchestrate the actions of various components, ensuring synchronization and coherence in the robot's responses. The adaptability of Python scripting allows for easy modifications, enabling the robot to evolve and incorporate new features seamlessly.

The flexibility of Python in scripting fosters a unified control language for diverse functionalities. Its dynamic nature allows for quick adjustments, ensuring that the e-waste robot can adapt to changing requirements and user expectations. Python's role in system control positions it as a versatile and indispensable tool in the ongoing development and enhancement of the e-waste robot.

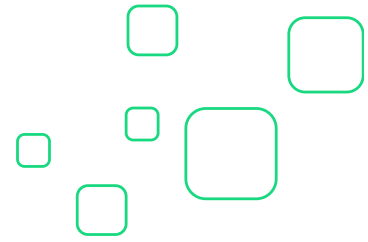


Conclusion:

In conclusion, the integration of Python in our e-waste robot project signifies a transformative leap in harnessing AI for system control. Python's versatility, combined with its powerful AI libraries, has been instrumental in realizing functionalities such as voice recognition, dynamic video display, and real-time voice conversation. The adaptability and scripting capabilities of Python position it as the backbone, infusing intelligence and interactivity into the e-waste robot. As we continue to explore the potential of Python and AI, the e-waste robot project stands as a testament to the possibilities of sustainable technology and innovative repurposing of electronic components.



Python's Speech Recognition Tools in the E-Waste Robot Project



Python's Rich Ecosystem:

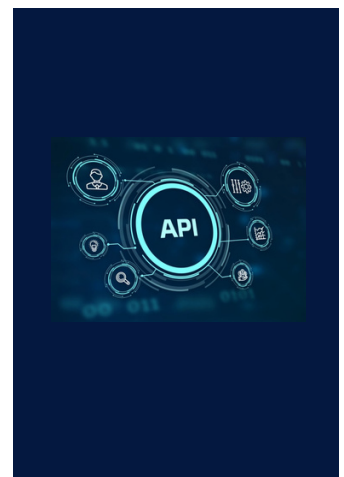
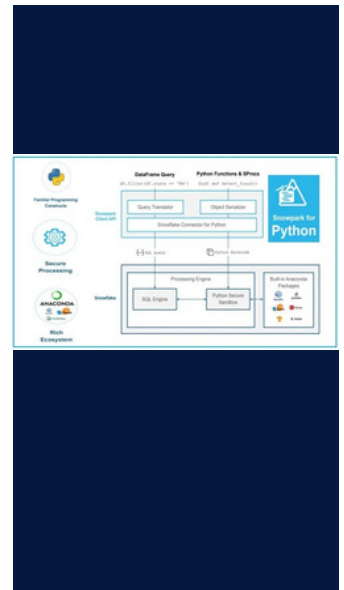
Python's extensive ecosystem of libraries includes powerful tools for speech recognition, a cornerstone of the e-waste robot project's interactive capabilities. This section delves into the specific speech recognition libraries employed, their integration into the overall system, and the advantages of using Python for handling voice inputs.

1. Speech Recognition Libraries Used:

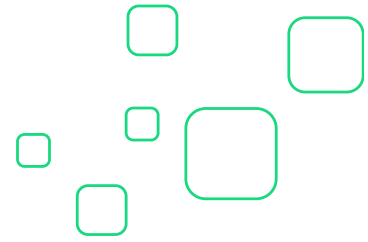
- 1.1 SpeechRecognition Library: The SpeechRecognition library in Python stands out for its simplicity and compatibility with multiple speech engines. It serves as a bridge to various engines, including Google Web Speech API, Sphinx, and Wit.ai. This flexibility allows the e-waste robot project to leverage different engines based on specific requirements.
- 1.2 Google Cloud Speech-to-Text API: Google Cloud Speech-to-Text API offers exceptional accuracy and supports multiple languages. By integrating this API through Python, the e-waste robot gains access to Google's robust cloud-based speech recognition capabilities. This choice aligns with the project's commitment to harnessing cutting-edge technologies.

2. Integration with the Overall System:

- 2.1 API Integration: Python's simplicity and versatility make it seamless to integrate speech recognition APIs into the overall system. The chosen libraries provide Python bindings or APIs that allow direct communication with the respective speech engines. This integration ensures that the e-waste robot can harness the power of these engines for accurate and efficient voice recognition.
- 2.2 Implementation with Python Scripting: Python scripting acts as the glue that binds the different components of the e-waste robot project. Using Python scripts, the integration of speech recognition libraries becomes a straightforward process. Scripts manage the flow of data between the speech recognition module, other AI components, and the robot's control system, ensuring a cohesive and synchronized operation.



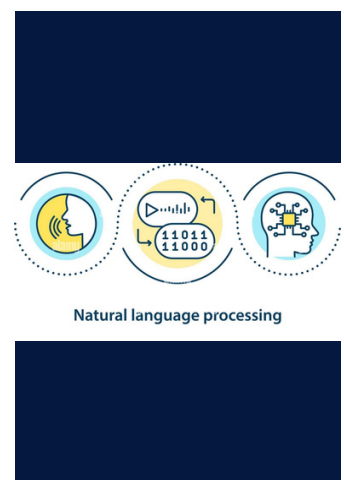
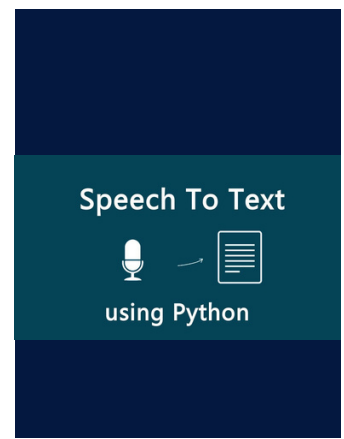
Python's Speech Recognition Tools in the E-Waste Robot Project



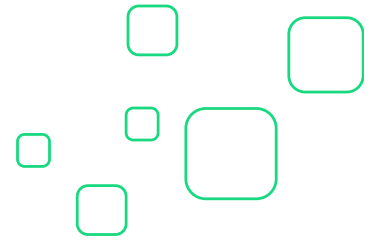
3. Advantages of Python in Handling Voice Inputs:

- 3.1 Flexibility in Data Handling: Python's flexibility in handling diverse data types is crucial for processing the varied nature of voice inputs. Whether it's raw audio data, transcribed text, or structured command outputs, Python provides the tools to efficiently manage and manipulate this data. This flexibility ensures that the e-waste robot can interpret and respond to voice commands with precision.
- 3.2 Integration with Natural Language Processing: Python's compatibility with natural language processing (NLP) enhances its capability to handle voice inputs intelligently. After converting spoken words into text, Python scripts can employ NLP techniques to extract meaning and context from the input. This enhances the e-waste robot's ability to understand user intent and execute appropriate actions.
- 3.3 Ease of Code Development: Python's clear and readable syntax simplifies code development for voice recognition. The straightforward implementation of logic for processing voice inputs contributes to the efficiency of the project. Python's readability also makes it easier for developers to collaborate on and maintain the codebase, a crucial aspect in the iterative development of the e-waste robot.

In conclusion, Python's rich ecosystem of speech recognition libraries, coupled with its seamless integration capabilities, plays a pivotal role in the success of the e-waste robot project. The chosen libraries, such as SpeechRecognition and Google Cloud Speech-to-Text API, are harnessed through Python scripting to enable accurate and versatile voice recognition. The advantages of Python in handling voice inputs, from its flexibility in data handling to its compatibility with natural language processing, position it as an ideal language for developing intelligent and responsive systems like the e-waste robot. As the project continues to evolve.



Python Scripts as Centralized Control Mechanism in the E-Waste Robot Project



3. Unified Scripting Language:

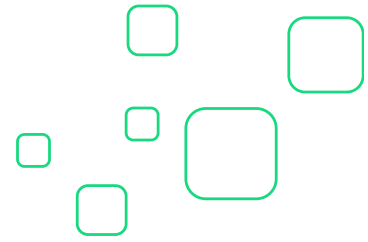
- 3.1 Cross-Component Compatibility: Python's versatility as a scripting language allows for the creation of unified scripts that are compatible across different components. This cross-component compatibility simplifies the integration process, as Python scripts can seamlessly interact with modules written for diverse functionalities. For example, a script managing voice recognition can easily communicate with a script controlling the robot's motion without encountering compatibility issues.
- 3.2 Code Reusability: Python's emphasis on readability and modularity promotes code reusability. This is particularly advantageous in the e-waste robot project, where functionalities such as voice recognition and video display may share common elements. Python scripts allow developers to encapsulate and reuse code segments efficiently, reducing redundancy and promoting a more maintainable codebase.
- 3.3 Adaptability to New Features: Python's dynamic nature facilitates adaptability, enabling scripts to evolve with the addition of new features. Whether integrating enhanced AI capabilities, incorporating additional sensors, or refining existing functionalities, Python scripts serve as flexible foundations that can be extended without major overhauls. This adaptability ensures that the e-waste robot can keep pace with evolving requirements.

Conclusion:

In conclusion, Python scripts act as the linchpin of the centralized control mechanism in the e-waste robot project, providing a unified scripting language that orchestrates actions and ensures synchronization. Their role in interfacing with diverse components, managing real-time decision-making, and offering a unified scripting language for cross-functionality positions Python as a cornerstone in the development and evolution of an intelligent and responsive robotic system.



Python Scripts as Centralized Control Mechanism in the E-Waste Robot Project



Centralized Control Architecture:

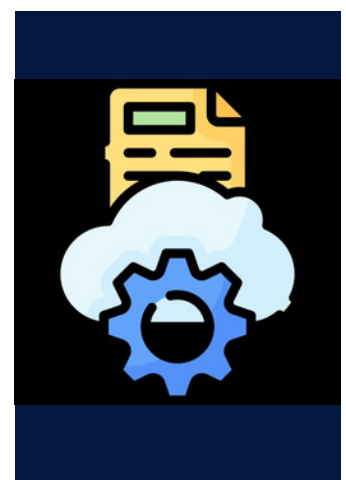
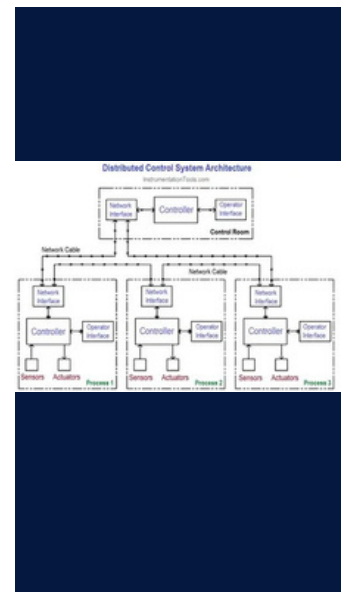
Python scripts play a pivotal role as the centralized control mechanism in the e-waste robot project, serving as the orchestration layer that harmonizes the actions of various components. This centralized control architecture ensures synchronization and coherence, enabling the robot to respond intelligently and seamlessly to user inputs and environmental stimuli.

1. Orchestrating Actions:

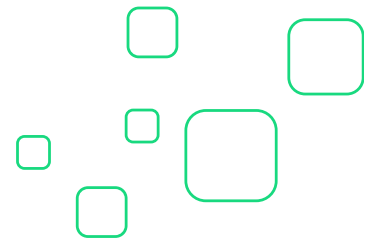
- 1.1 Interfacing with Components: Python scripts act as intermediaries, interfacing with different components of the e-waste robot. These components may include speech recognition modules, computer vision systems, motors, actuators, and other electronic devices. By orchestrating communication between these diverse components, Python scripts enable a coordinated and integrated response to external stimuli.
- 1.2 Real-time Decision Making: The real-time nature of Python scripting allows for dynamic decision-making based on incoming data. Whether it's processing voice commands, analyzing visual inputs, or responding to sensor readings, Python scripts make split-second decisions, contributing to the overall responsiveness and interactivity of the e-waste robot.

2. Ensuring Synchronization:

- 2.1 Time-Stamped Execution: Python scripts are designed to execute commands in a time-stamped manner, ensuring synchronization between different actions. This temporal coordination is critical, especially in scenarios where multiple components need to act in harmony, such as during voice recognition, video display, and motion control. Python's built-in functionalities for time management contribute to the precision and coherence of the robot's responses.
- 2.2 Event-Driven Architecture: Python scripts often adopt an event-driven architecture, responding to specific triggers or stimuli. Events can include user commands, changes in the environment, or inputs from various sensors.



CODE – Artificial Intelligence



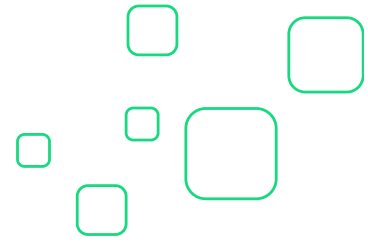
```
import time
import pygame
import speech_recognition as sr
import datetime
import wikipedia
import os
import subprocess
import random
import pywhatkit
import webbrowser
# from selenium import webdriver
# from selenium.webdriver.common.by import By
# from selenium.webdriver.chrome.options import Options
import psutil
import vlc
# from transformers import pipeline
import threading
import requests
import tkinter as tk
```

```
# ENTER_TEXT = "[id='__next'] .grow.overflow-x-auto .relative.flex.h-full.w-full textarea";
# BUTTON_ICON = "[id='__next'] .grow.overflow-x-auto .relative.flex.h-full.w-full button";
# REPLY_TEXT = "[id='__next'] [class='py-1 relative break-anywhere'] span";
```

```
# def getReplyFromPi(question):
#     driver = webdriver.Chrome()
#     # chrome_options = Options()8
#     # chrome_options.add_argument('--log-level=3')
#     # chrome_options.add_argument("--headless")
#     # driver = webdriver.Chrome(options=chrome_options)
#     driver.get("https://pi.ai/home")
#     time.sleep(5)
if driver.current_url == "https://pi.ai/onboarding":
#     print('ghkkfghgfh')
#     driver.get("https://pi.ai/home")
#     # driver.headless = True
#     # driver.maximize_window()
#     i = 1
#     time.sleep(2)
```



CODE – Artificial Intelligence



```
# driver.find_element(By.CSS_SELECTOR, ENTER_TEXT).send_keys(question)
# time.sleep(2)
# driver.find_element(By.CSS_SELECTOR, BUTTON_ICON).click()
# time.sleep(2)
# ele = driver.find_elements(By.CSS_SELECTOR, REPLY_TEXT)
# return ele[i+0].text
def vlc_video(video_path):
    # Initialize the VLC instance
    instance = vlc.Instance()
    player = instance.media_player_new()

    # Load the video file
    media = instance.media_new(video_path)
    player.set_media(media)

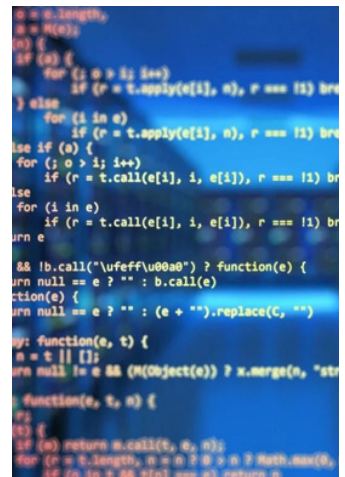
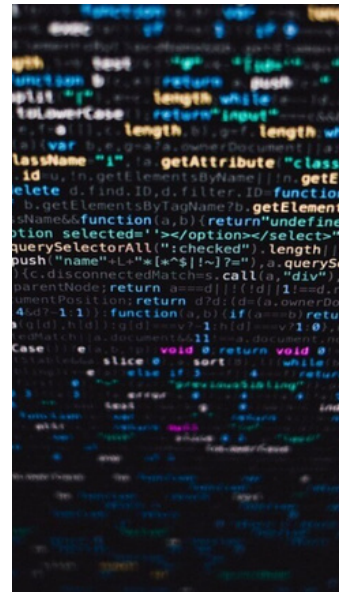
    # Play the video
    player.play()

def speak(data):
    voice = 'en-CA-LiamNeural'
    time_label.config(text="Ethan : "+ data)
    command = f'edge-tts --voice "{voice}" --text "{data}" --write-media "data.mp3"'
    os.system(command)

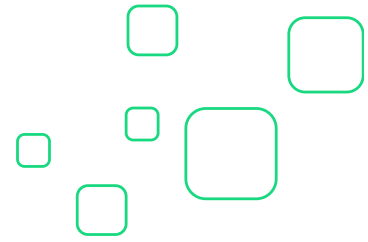
    pygame.init()
    pygame.mixer.init()
    pygame.mixer.music.load("data.mp3")
    try:
        pygame.mixer.music.play()

        while pygame.mixer.music.get_busy():
            pygame.time.Clock().tick(10)

    except Exception as e:
        print(e)
    finally:
        pygame.mixer.music.stop()
        pygame.mixer.quit()
```



CODE – Artificial Intelligence



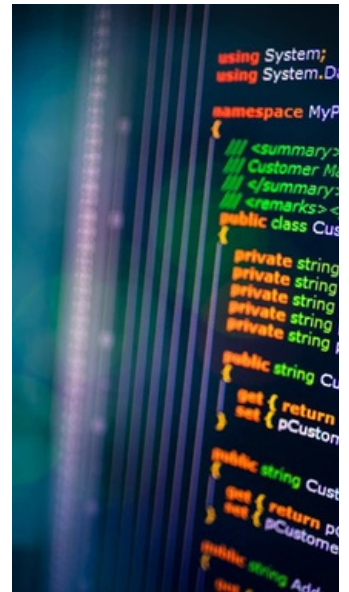
```
def play_music():
    global music_playing
    music_dir = 'C:\\E3N' # Update with your music directory
    songs = os.listdir(music_dir)
    if len(songs) > 0:
        os.startfile(os.path.join(music_dir, songs[0]))
        music_playing = True
        speak('Playing music.')
    else:
        speak('No music found in the directory.')

def searchGoogle(query):
    if "google" in query:
        import wikipedia as googleScrap
        query = query.replace("ethan", "")
        query = query.replace("e3n", "")
        query = query.replace("ethane", "")
        query = query.replace("google search", "")
        query = query.replace("google", "")
        speak("This is what I found on google")

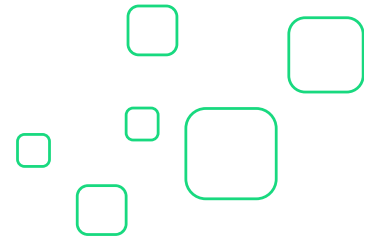
        try:
            pywhatkit.search(query)
            result = googleScrap.summary(query, 1)
            speak(result)
        except:
            speak("No speakable output available")

def searchYoutube(query):
    if "youtube" in query:
        speak("This is what I found for your search!")
        query = query.replace("youtube search", "")
        query = query.replace("youtube", "")
        query = query.replace("jarvis", "")
        web = "https://www.youtube.com/results?search_query=" + query
        webbrowser.open(web)
        pywhatkit.playonyt(query)
        speak("Done, Sir")

def shutdown_computer():
    try:
        os.system("shutdown /s /t 0") # Initiates a shutdown immediately (/s: shutdown, /t:
time, 0: immediately)
    except Exception as e:
        print(f"An error occurred: {str(e)}")
```



CODE – Artificial Intelligence

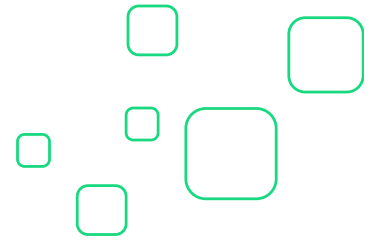


```
def game_play():
    speak("Lets Play ROCK PAPER SCISSORS !!")
    print("LETS PLAYYYYYYYYYYYYYYYY")
    i = 0
    Me_score = 0
    Com_score = 0
    while(i<5):
        choose = ("rock","paper","scissors") #Tuple
        com_choose = random.choice(choose)
        query = takeCommand().lower()
        if (query == "rock"):
            if (com_choose == "rock"):
                speak("ROCK")
                print(f"Score:- ME :- {Me_score} : COM :- {Com_score}")
            elif (com_choose == "paper"):
                speak("paper")
                Com_score += 1
                print(f"Score:- ME :- {Me_score} : COM :- {Com_score}")
            else:
                speak("Scissors")
                Me_score += 1
                print(f"Score:- ME :- {Me_score} : COM :- {Com_score}")

        elif (query == "paper" ):
            if (com_choose == "rock"):
                speak("ROCK")
                Me_score += 1
                print(f"Score:- ME :- {Me_score+1} : COM :- {Com_score}")

            elif (com_choose == "paper"):
                speak("paper")
                print(f"Score:- ME :- {Me_score} : COM :- {Com_score}")
            else:
                speak("Scissors")
                Com_score += 1
                print(f"Score:- ME :- {Me_score} : COM :- {Com_score}")
        elif(query == "exit" or query == "quit"):
            break
        elif (query == "scissors" or query == "scissor"):
            if (com_choose == "rock"):
                speak("ROCK")
                Com_score += 1
                print(f"Score:- ME :- {Me_score} : COM :- {Com_score}")
            elif (com_choose == "paper"):
                speak("paper")
                Me_score += 1
                print(f"Score:- ME :- {Me_score} : COM :- {Com_score}")
```

CODE – Artificial Intelligence



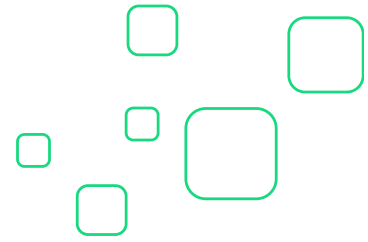
```
else:
    speak("Scissors")
    print(f"Score:- ME :- {Me_score} : COM :- {Com_score}")
    i += 1

print(f"FINAL SCORE :- ME :- {Me_score} : COM :- {Com_score}")
```

```
def wishMe():
    hour = datetime.datetime.now().hour
    if 0 <= hour < 12:
        speak(f"Good Morning!")
        time_label.config(text="Good Morning!")
        speak("Hello, I am E 3 N, an evolution human version 3 robot. I have been created for
war. I met one of my best captain rayes who died in front of me. Well, I think I could change my
past but here I am okay and fine with my fellow friend. Shubh I'm glad that we are good friends.
lets make a conversation")
    elif 12 <= hour < 18:
        speak(f"Good Afternoon!")
        time_label.config(text="Good Afternoon!")
        speak("Hello, I am E 3 N, an evolution human version 3 robot. I have been created for
war. I met one of my best captain rayes who died in front of me. Well, I think I could change my
past but here I am okay and fine with my fellow friend. Shubh I'm glad that we are good friends.
lets make a conversation")
    else:
        speak(f"Good Evening!")
        time_label.config(text="Good Evening!")
        speak("Hello, I am E 3 N, an evolution human version 3 robot. I have been created for
war. I met one of my best captain rayes who died in front of me. Well, I think I could change my
past but here I am okay and fine with my fellow friend. Shubh I'm glad that we are good friends.
lets make a conversation")
def takeCommand():
    r = sr.Recognizer()
    with sr.Microphone() as source:
        speak("Listening...")
        r.pause_threshold = 1
        audio = r.listen(source)

    try:
        speak("Recognizing...")
        query = r.recognize_google(audio, language='en-us')
        print(f"User said: {query}\n")
        time_label.config(text="You : " + query)
        time.sleep(3)
```

CODE – Artificial Intelligence



```
except Exception as e:
```

```
    time_label.config(text="Say that again, please...")
```

```
    speak("Say that again, please...")
```

```
    return "None"
```

```
    return query
```

```
API_URL = "https://api-inference.huggingface.co/models/facebook/blenderbot-400M-distill"
```

```
headers = {"Authorization": "Bearer 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"}
```

```
def ai(input_text):
```

```
    payload = {
```

```
        "inputs": {
```

```
            "text": input_text
```

```
        }
```

```
    }
```

```
    response = requests.post(API_URL, headers=headers, json=payload)
```

```
    return response.json()
```

```
def over():
```

```
    video_folder = "C:\\E3N\\media\\else"
```

```
    video_pa = 'C:\\Program Files\\VideoLAN\\VLC\\vlc.exe'
```

```
# Get a list of all video files in the folder
```

```
video_files = [f for f in os.listdir(video_folder) if f.endswith((".mp4", ".avi", ".mkv"))]
```

```
if not video_files:
```

```
    print("No video files found in the folder.")
```

```
else:
```

```
    # Choose a random video file
```

```
    random_video = random.choice(video_files)
```

```
    video_path = os.path.join(video_folder, random_video)
```

```
# Command to open VLC in fullscreen mode and play the selected video
```

```
vlc_command = [video_pa, '--fullscreen', video_path]
```

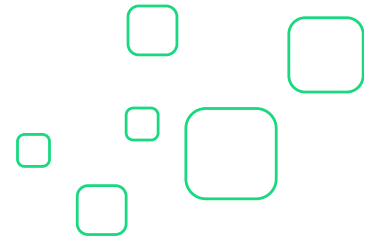
```
try:
```

```
    # Open VLC and play the video
```

```
    subprocess.Popen(vlc_command)
```

```
    print(f"Playing video: {random_video}")
```


CODE – Artificial Intelligence



```
# Wait for the video to play for 25 seconds
time.sleep(50)

# Stop the video playback
subprocess.Popen([video_pa, '--stop'])

except FileNotFoundError:
    print("VLC player not found. Make sure VLC is installed and in your system's PATH.")

def me():
    video_folder = "C:\\E3N\\media\\else\\me"
    video_pa = 'C:\\Program Files\\VideoLAN\\VLC\\vlc.exe'

    # Get a list of all video files in the folder
    video_files = [f for f in os.listdir(video_folder) if f.endswith((".mp4", ".avi", ".mkv"))]

    if not video_files:
        print("No video files found in the folder.")
    else:
        # Choose a random video file
        random_video = random.choice(video_files)
        video_path = os.path.join(video_folder, random_video)

    # Command to open VLC in fullscreen mode and play the selected video
    vlc_command = [video_pa, '--fullscreen', video_path]

    try:
        # Open VLC and play the video
        subprocess.Popen(vlc_command)
        print(f"Playing video: {random_video}")

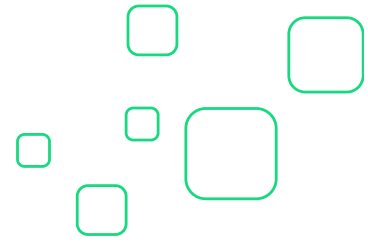
        # Wait for the video to play for 25 seconds
        time.sleep(25)

        # Stop the video playback
        subprocess.Popen([video_pa, '--stop'])

    except FileNotFoundError:
        print("VLC player not found. Make sure VLC is installed and in your system's PATH.")

def nolan():
    video_folder = "C:\\E3N\\media\\else\\nolan"
    video_pa = 'C:\\Program Files\\VideoLAN\\VLC\\vlc.exe'
```

CODE – Artificial Intelligence



```
# Get a list of all video files in the folder
video_files = [f for f in os.listdir(video_folder) if f.endswith((".mp4", ".avi", ".mkv"))]

if not video_files:
    print("No video files found in the folder.")
else:
    # Choose a random video file
    random_video = random.choice(video_files)
    video_path = os.path.join(video_folder, random_video)

    # Command to open VLC in fullscreen mode and play the selected video
    vlc_command = [video_pa, '--fullscreen', video_path]

    try:
        # Open VLC and play the video
        subprocess.Popen(vlc_command)
        print(f"Playing video: {random_video}")

        # Wait for the video to play for 25 seconds
        time.sleep(25)

        # Stop the video playback
        subprocess.Popen([video_pa, '--stop'])
    except FileNotFoundError:
        print("VLC player not found. Make sure VLC is installed and in your system's PATH.")

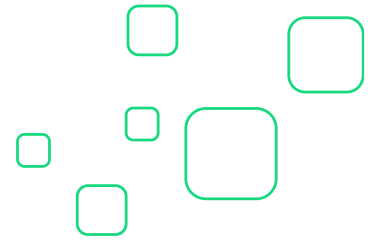
def pain():
    video_folder = "C:\\E3N\\media\\else\\pain"
    video_pa = 'C:\\Program Files\\VideoLAN\\VLC\\vlc.exe'

    # Get a list of all video files in the folder
    video_files = [f for f in os.listdir(video_folder) if f.endswith((".mp4", ".avi", ".mkv"))]

    if not video_files:
        print("No video files found in the folder.")
    else:
        # Choose a random video file
        random_video = random.choice(video_files)
        video_path = os.path.join(video_folder, random_video)

        # Command to open VLC in fullscreen mode and play the selected video
        vlc_command = [video_pa, '--fullscreen', video_path]
```

CODE – Artificial Intelligence



```
try:
    # Open VLC and play the video
    subprocess.Popen(vlc_command)
    print(f"Playing video: {random_video}")

    # Wait for the video to play for 25 seconds
    time.sleep(25)

    # Stop the video playback
    subprocess.Popen([video_pa, '--stop'])

except FileNotFoundError:
    print("VLC player not found. Make sure VLC is installed and in your system's PATH.")
def song():
    video_folder = "C:\\E3N\\media\\else\\song"
    video_pa = 'C:\\Program Files\\VideoLAN\\VLC\\vlc.exe'

    # Get a list of all video files in the folder
    video_files = [f for f in os.listdir(video_folder) if f.endswith((".mp4", ".avi", ".mkv"))]

    if not video_files:
        print("No video files found in the folder.")
    else:
        # Choose a random video file
        random_video = random.choice(video_files)
        video_path = os.path.join(video_folder, random_video)

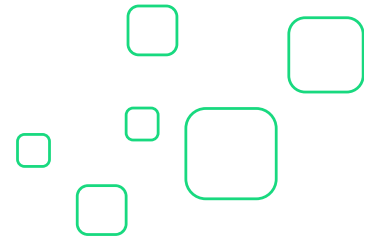
    # Command to open VLC in fullscreen mode and play the selected video
    vlc_command = [video_pa, '--fullscreen', video_path]

    try:
        # Open VLC and play the video
        subprocess.Popen(vlc_command)
        print(f"Playing video: {random_video}")

        # Wait for the video to play for 25 seconds
        time.sleep(25)

        # Stop the video playback
        subprocess.Popen([video_pa, '--stop'])
```

CODE – Artificial Intelligence



```
except FileNotFoundError:
    print("VLC player not found. Make sure VLC is installed and in your system's PATH.")
def world():
    video_folder = "C:\\E3N\\media\\else\\world"
    video_pa = 'C:\\Program Files\\VideoLAN\\VLC\\vlc.exe'

    # Get a list of all video files in the folder
    video_files = [f for f in os.listdir(video_folder) if f.endswith((".mp4", ".avi", ".mkv"))]

    if not video_files:
        print("No video files found in the folder.")
    else:
        # Choose a random video file
        random_video = random.choice(video_files)
        video_path = os.path.join(video_folder, random_video)

        # Command to open VLC in fullscreen mode and play the selected video
        vlc_command = [video_pa, '--fullscreen', video_path]

        try:
            # Open VLC and play the video
            subprocess.Popen(vlc_command)
            print(f"Playing video: {random_video}")

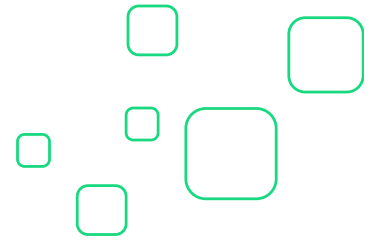
            # Wait for the video to play for 25 seconds
            time.sleep(25)
        # Stop the video playback
        subprocess.Popen([video_pa, '--stop'])

    except FileNotFoundError:
        print("VLC player not found. Make sure VLC is installed and in your system's PATH.")

def baldev():
    video_folder = "C:\\E3N\\media\\shivam"
    video_pa = 'C:\\Program Files\\VideoLAN\\VLC\\vlc.exe'

    # Get a list of all video files in the folder
    video_files = [f for f in os.listdir(video_folder) if f.endswith((".mp4", ".avi", ".mkv"))]
```

CODE – Artificial Intelligence



```

    elif not video_files:
        print("No video files found in the folder.")
    else:
        # Choose a random video file
        random_video = random.choice(video_files)
        video_path = os.path.join(video_folder, random_video)

        # Command to open VLC in fullscreen mode and play the selected video
        vlc_command = [video_pa, '--fullscreen', video_path]

        try:
            # Open VLC and play the video
            subprocess.Popen(vlc_command)
            print(f"Playing video: {random_video}")

            # Wait for the video to play for 25 seconds
            time.sleep(25)

            # Stop the video playback
            subprocess.Popen([video_pa, '--stop'])

        except FileNotFoundError:
            print("VLC player not found. Make sure VLC is installed and in your system's PATH.")

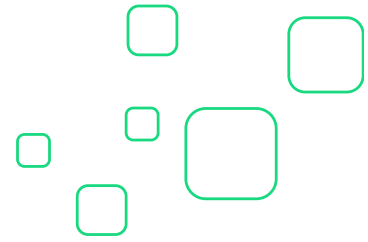
def fft():
    subprocess.Popen(['python', 'C:\\E3N\\spectrum\\Realtime_PyAudio_FFT-master\\run_FFT_analyzer.py'])

def comm():
    music_playing = False # Initialize music playing status
    wishMe()
    fft()
    video_pa = 'C:\\Program Files\\VideoLAN\\VLC\\vlc.exe'
    while True:
        query = takeCommand().lower()

        if 'ethane' in query:
            query = query.replace("ethane", "")
            speak('let me think')
            results = wikipedia.summary(query, sentences=2)
            speak("its easy dude")
            speak(results)
        elif query=="kakashi":
            video_path = 'C:\\E3N\\media\\Apoorva.mp4'
            subprocess.Popen([video_pa, '--fullscreen', video_path])
            time.sleep(9)
            stop()

```

CODE – Artificial Intelligence



```
elif "play something" in query :
```

```
    over()
```

```
    stop()
```

```
elif 'baldev the destroyer' in query:
```

```
    baldev()
```

```
    stop()
```

```
elif 'subhajeet sen' in query:
```

```
    video_path = 'C:\\E3N\\media\\subhajit.mp4'
```

```
    subprocess.Popen([video_pa, '--fullscreen', video_path])
```

```
    time.sleep(16)
```

```
    stop()
```

```
elif 'mind sync' in query:
```

```
    video_path = 'C:\\E3N\\media\\sourav.mp4'
```

```
    subprocess.Popen([video_pa, '--fullscreen', video_path])
```

```
    time.sleep(10)
```

```
    stop()
```

```
elif 'itachi' in query:
```

```
    video_path = 'C:\\E3N\\media\\GOURAB.mp4'
```

```
    subprocess.Popen([video_pa, '--fullscreen', video_path])
```

```
    time.sleep(18)
```

```
    stop()
```

```
elif 'live life' in query:
```

```
    video_path = 'C:\\E3N\\media\\tina.mp4'
```

```
    subprocess.Popen([video_pa, '--fullscreen', video_path])
```

```
    time.sleep(25)
```

```
    stop()
```

```
elif 'spidey' in query:
```

```
    video_path = 'C:\\E3N\\media\\Suranjan.mp4'
```

```
    subprocess.Popen([video_pa, '--fullscreen', video_path])
```

```
    time.sleep(14)
```

```
    stop()
```

```
elif 'royal princess' in query:
```

```
    video_path = 'C:\\E3N\\media\\Nishu.mp4'
```

```
    subprocess.Popen([video_pa, '--fullscreen', video_path])
```

```
    time.sleep(32)
```

```
    stop()
```

```
elif 'jeet karmakar' in query:
```

```
    video_path = 'C:\\E3N\\media\\jeet.mp4'
```

```
    subprocess.Popen([video_pa, '--fullscreen', video_path])
```

```
    time.sleep(13)
```

```
    stop()
```

```
elif 'wonder woman' in query:
```

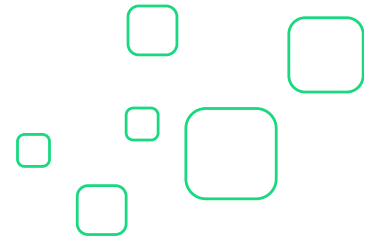
```
    video_path = 'C:\\E3N\\media\\nikita.mp4'
```

```
    subprocess.Popen([video_pa, '--fullscreen', video_path])
```

```
    time.sleep(18)
```

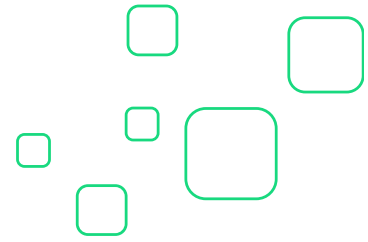
```
    stop()
```


CODE – Artificial Intelligence



```
elif 'gear fifth' in query:
    video_path = 'C:\\E3N\\media\\Srijon.mp4'
    subprocess.Popen([video_pa, '--fullscreen', video_path])
    time.sleep(14)
    stop()
elif 'infinity x' in query:
    video_path = 'C:\\E3N\\media\\tomo.mp4'
    subprocess.Popen([video_pa, '--fullscreen', video_path])
    time.sleep(20)
    stop()
elif 'paradise point' in query:
    video_path = 'C:\\E3N\\media\\ritika.mp4'
    subprocess.Popen([video_pa, '--fullscreen', video_path])
    time.sleep(31)
    stop()
elif 'into the heavens' in query:
    video_path = 'C:\\E3N\\media\\aqdas.mkv'
    subprocess.Popen([video_pa, '--fullscreen', video_path])
    time.sleep(11)
    stop()
elif 'technova' in query:
    video_path = 'C:\\E3N\\media\\sujoy.mp4'
    subprocess.Popen([video_pa, '--fullscreen', video_path])
    time.sleep(20)
    stop()
elif 'superhero' in query:
    video_path = 'C:\\E3N\\media\\rounak.mp4'
    subprocess.Popen([video_pa, '--fullscreen', video_path])
    time.sleep(10)
    stop()
elif 'shadow clone' in query:
    video_path = 'C:\\E3N\\media\\aradhya.mp4'
    subprocess.Popen([video_pa, '--fullscreen', video_path])
    time.sleep(12)
    stop()
elif 'meaning of life' in query:
    video_path = 'C:\\E3N\\media\\life (1).mp4'
    subprocess.Popen([video_pa, '--fullscreen', video_path])
    time.sleep(165)
    stop()
elif 'best feelings in the world' in query:
    video_path = 'C:\\E3N\\media\\life (2).mp4'
    subprocess.Popen([video_pa, '--fullscreen', video_path])
    time.sleep(38)
    stop()
```

CODE – Artificial Intelligence



elif 'take me to the space' in query:

```
video_path = 'C:\\E3N\\media\\life (3).mp4'
subprocess.Popen([video_pa, '--fullscreen', video_path])
time.sleep(28)
stop()
```

elif 'give me advice' in query:

```
video_path = 'C:\\E3N\\media\\life (4).mp4'
subprocess.Popen([video_pa, '--fullscreen', video_path])
time.sleep(79)
stop()
```

elif 'beauty of universe' in query:

```
video_path = 'C:\\E3N\\media\\life (5).mp4'
subprocess.Popen([video_pa, '--fullscreen', video_path])
time.sleep(20)
stop()
```

elif 'meaning of failure' in query:

```
video_path = 'C:\\E3N\\media\\life (6).mp4'
subprocess.Popen([video_pa, '--fullscreen', video_path])
time.sleep(20)
stop()
```

elif 'dream of universe' in query:

```
video_path = 'C:\\E3N\\media\\life (7).mp4'
subprocess.Popen([video_pa, '--fullscreen', video_path])
time.sleep(18)
stop()
```

elif 'what joker said' in query:

```
video_path = 'C:\\E3N\\media\\life (8).mp4'
subprocess.Popen([video_pa, '--fullscreen', video_path])
time.sleep(21)
stop()
```

elif 'captain of the ship' in query:

```
video_path = 'C:\\E3N\\media\\shubh.mp4'
subprocess.Popen([video_pa, '--fullscreen', video_path])
time.sleep(17)
speak("master shubh will be a best pirates in the world ")
stop()
```

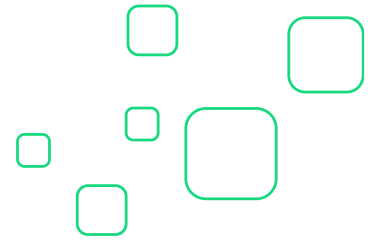
elif 'something sometimes' in query:

```
video_path = 'C:\\E3N\\media\\soul.mp4'
subprocess.Popen([video_pa, '--fullscreen', video_path])
time.sleep(16)
stop()
```

elif 'long live the king' in query:

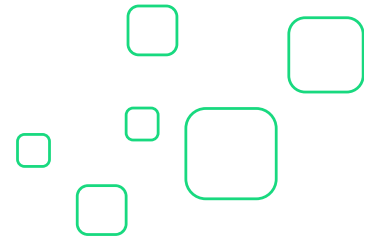
```
video_path = 'C:\\E3N\\media\\inter.mp4'
subprocess.Popen([video_pa, '--fullscreen', video_path])
time.sleep(7)
stop()
```

CODE – Artificial Intelligence



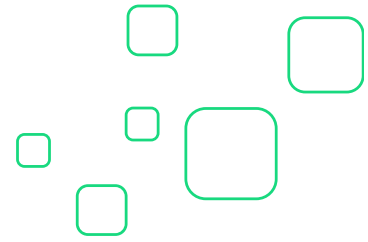
```
elif 'blade runner' in query:
    video_path = 'C:\\E3N\\media\\runner.mp4'
    subprocess.Popen([video_pa, '--fullscreen', video_path])
    time.sleep(25)
    stop()
elif 'get lost' in query:
    speak("good bye master shubh , i will say at last you are the best one in the
world")
    shutdown_computer()
else:
    speak(f'Neural conversation mode activated. now you can ask me anything. just
say hello or amigo")
    while True:
        query = takeCommand().lower()
        if 'ethane' in query:
            query = query.replace("ethane", "")
            speak('let me think')
            results = wikipedia.summary(query, sentences=2)
            speak(" its easy dude")
            speak(results)
        elif "google" in query:
            searchGoogle(query)
        elif "close chrome" in query:
            speak("closing chrome sir just wait a minute")
            kill_chrome()
        elif 'exit' in query or "admin" in query:
            speak("admin mode has been activated")
            break
        elif " close chrome" in query:
            speak("closing chrome sir just wait a minute")
            kill_chrome()
        elif "youtube" in query:
            searchYoutube(query)
    elif "wikipedia" in query:
        searchWikipedia(query)
    elif 'play game' in query:
        game_play()
    elif 'hello' in query or 'hey' in query or 'hi' in query:
        speak(f'hi . how are you ?')
    elif 'i am fine' in query or 'i m fine' in query:
        speak(f' hi . i am glad that you are well . if you want to know me. just say .
```

CODE – Artificial Intelligence



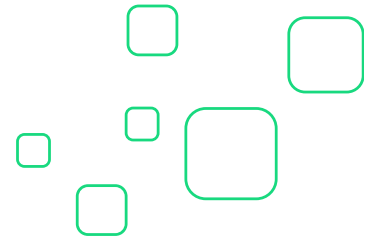
```
elif 'i am fine' in query or 'i m fine' in query:
    speak(f' hi . i am glad that you are well . if you want to know me. just say .
tell me about yourself . or have questions say e3n or amigo')
elif 'tell me about yourself' in query:
    speak(
        f'Ethan, an advanced AI named E3N, was an invaluable ally to the UNSA
Retribution in the war against the SDF. His friendship with Captain Reyes and the crew
grew as they faced relentless battles across the Solar System. Ethan's wit and
resourcefulness saved them more times than they could count. One day, amidst a fierce
battle, the Retribution found itself stranded in space, damaged and struggling. It was
Ethan's moment to shine. With his unmatched intellect, he guided the crew, making
crucial repairs and devising a plan to survive. As they held on to hope, Ethan remained
their steadfast companion, proving that bonds of friendship and the power of resilience
could conquer even the darkest of challenges in the vastness of space. for next . say .
who created you .")
elif 'who created you' in query:
    speak(
        f'mister shubh kumar sinha has created in a lab with limited resources and
guided by proffessor sujeet goswami sir . do you want to know about shoev kumar . say
. tell me about master')
elif 'tell me about master' in query:
    speak(
        f'well shubh kumar worked with its team matrix and performing myself to
you . he is just a normal person but he not live in this real world . he believed that
escaping from reality is the good way to create your own world where you are the only
king of that world. for my story video say . play your story')
elif 'play your story' in query:
    video_path = 'C:\\E3N\\media\\Ethanel.mp4'
    subprocess.Popen([video_pa, '--fullscreen', video_path])
    time.sleep(332)
    stop()
    speak('if you want to know about team member . say . team matrix')
elif 'team matrix' in query:
    time.sleep(1)
    speak(f'in this project. shubh kumar leads this team where apoorav pandit.
shivam kumar. tina roy. tomojeet hazra. subhajeet sen. akdas sultan. jeet karmakar.
gourab kundu. suranjan sen gupta. srijon. and sabhyasachi bose. from second year as
well as from third year nikita yadav. sujoy ghosal. rakesh kumar chouhan. ritika das.
nishu kumari. rounak gupta and sourav bhattacharya. worked together as a team . if you
want to ask anything . just say my name such as e3n or ethan and then your question ,
like. ethan what is dinosaur')
```

CODE – Artificial Intelligence



```
elif 'time' in query:
    strTime = datetime.datetime.now().strftime("%H:%M:%S")
    speak(f"Sir, the time is {strTime}")

elif 'black hole' in query:
    video_path = 'C:\\E3N\\media\\else\\black.mp4'
    subprocess.Popen([video_pa, '--fullscreen', video_path])
    time.sleep(18)
    stop()
elif 'love' in query:
    video_path = 'C:\\E3N\\media\\else\\love.mp4'
    subprocess.Popen([video_pa, '--fullscreen', video_path])
    time.sleep(18)
    stop()
elif 'song' in query or 'songs' in query:
    song()
    stop()
elif 'nolan' in query:
    nolan()
    stop()
elif 'painful thoughts' in query:
    pain()
    stop()
elif 'creator' in query:
    me()
    stop()
elif 'desire' in query:
    world()
    stop()
elif "amigo" in query or "e3n" in query:
    speak("ask your question sir")
    while True:
        query = takeCommand()
        if "quit" in query or "exit" in query:
            speak("going back to normal mode")
            break
        else:
            output = ai(query)
            sp = output['generated_text']
            speak(sp)
```



Conclusion –

The culmination of the E-Waste Robot Project marks not just the completion of a comprehensive technological endeavor but a testament to the transformative power of innovation, sustainability, and collaborative effort. Throughout this project, our team delved into the intricate realms of repurposing electronic waste, blending diverse technical disciplines to bring forth the marvel that is E3AN.

From the project's inception, our primary objective was to repurpose non-functioning computers into a functional and aesthetically pleasing robot. We envisioned a sustainable solution that not only addressed the growing issue of e-waste but also showcased the potential for creative and environmentally conscious technology. As we reflect on the journey, it is evident that we have not only met but exceeded our initial goals.

The technical backbone of the project rests on the intricacies of 3D design and crafting. Utilizing AutoCAD for precise schematics and Pepakura for the outer covering allowed us to breathe life into our envisioned robot. The incorporation of mechanical work, including welding, metal cutting, and shaping, further solidified the physical manifestation of E3AN, transforming discarded components into a cohesive and functional robotic entity.

The integration of IoT components and artificial intelligence stands as a hallmark of our project. Arduino served as the brain, orchestrating the diverse components seamlessly. Electronic devices, including motor drivers, ultrasonic sensors, LED lights, and servo motors, not only contributed to the functionality but also elevated E3AN to an interactive and intelligent system. Python, our chosen language for AI applications, facilitated voice recognition, dynamic video display, and real-time voice conversations, making E3AN a dynamic and engaging creation.

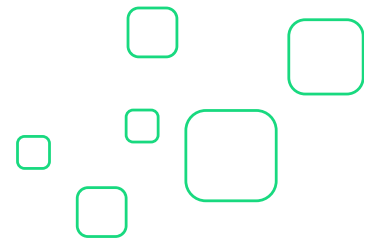
Certainly, the journey was not without its challenges. From technical intricacies to project management complexities, each hurdle provided an opportunity for growth and learning. The collaborative spirit of our team, combined with the guidance of our mentors, proved instrumental in overcoming obstacles and pushing the boundaries of our collective knowledge.

As we stand at the conclusion of this project, we do not view it as an endpoint but rather as a launchpad for future endeavors. The success of E3AN propels us to explore further innovations, delve into new technologies, and continue our commitment to sustainable and responsible technology development. We envision E3AN as a harbinger of change, inspiring future projects that fuse technological brilliance with environmental consciousness.

In closing, we extend our heartfelt gratitude to all those who supported and guided us throughout this journey. Our mentors, faculty advisors, and peers have been indispensable in shaping the success of the E-Waste Robot Project. Their expertise, encouragement, and constructive feedback have been invaluable.

This project report stands as a testament to our dedication, creativity, and the limitless possibilities that arise when a group of passionate individuals comes together with a shared vision. As we bid farewell to the E-Waste Robot Project, we carry forward the knowledge, experiences, and inspiration gained, confident that the impact of our efforts will reverberate far beyond the confines of this academic endeavor.

Future Scope



The culmination of the E-Waste Robot Project marks not just the completion of a comprehensive technological endeavor but a testament to the transformative power of innovation, sustainability, and collaborative effort. Throughout this project, our team delved into the intricate realms of repurposing electronic waste, blending diverse technical disciplines to bring forth the marvel that is E3AN.

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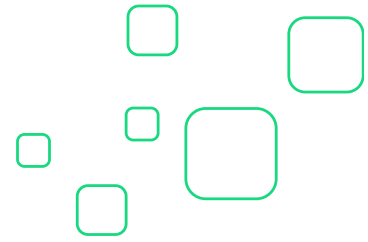
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- [7] <https://github.com/topics/python-ai>
- [8] <https://codesubmit.io/blog/ai-code-tools/>
- [9] <https://artint.info/AIPython/aipython/aipython>
- [10] <https://www.geeksforgeeks.org/build-a-virtual-assistant-using-python/>



Thank You